Notebook ICPC
La Tasita

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Dynamic Programming

Convex Hull Trick

```
// Convex Hull Trick (CHT) - O(1) amortizado por insercion y
query
// Optimiza: dp[i] = min/max ( m_j * x_i + c_j )
// Condicion: Pendientes 'm' insertadas en orden MONOTONO
(creciente/decreciente).
// Para queries 'x' monotonos: query() usa dos punteros O(1).
// Para queries 'x' arbitrarios: query_bin() usa busqueda
binaria O(log N).

struct Line {
    ll m, c;
    ll eval(ll x) const { return m * x + c; }
};

// CHT para MINIMIZAR (m decreciente, queries x crecientes)
// Para MAXIMIZAR: invertir signos o usar m creciente y
cambiar condicion is_bad
struct CHT {
    vector<Line> hull;
    int ptr = 0;

    // Retorna true si l2 es redundante entre l1 y l3
    bool is_bad(const Line& l1, const Line& l2, const Line&
l3) {
        // Interseccion(l1, l2) >= Interseccion(l2, l3) usando
__int128 para evitar overflow
        return (__int128)(l1.c - l2.c) * (l3.m - l2.m) >=
(__int128)(l2.c - l3.c) * (l2.m - l1.m);
    }

    void add(ll m, ll c) {
        Line l = {m, c};
        while (sz(hull) >= 2 && is_bad(hull[sz(hull) - 2],
hull.back(), l)) {
            hull.pop_back();
        }
        hull.pb(l);
        if (ptr >= sz(hull)) ptr = sz(hull) - 1;
    }
```

```
    // Query O(1) amortizado (requiere x monotonamente
creciente)
    ll query(ll x) {
        if (hull.empty()) return LINF;
        while (ptr + 1 < sz(hull) && hull[ptr + 1].eval(x) <=
hull[ptr].eval(x)) {
            ++ptr;
        }
        return hull[ptr].eval(x);
    }

    // Query O(log N) para x arbitrarios (no requiere que x
sea monotono)
    ll query_bin(ll x) {
        if (hull.empty()) return LINF;
        int l = 0, r = sz(hull) - 1;
        while (l < r) {
            int mid = l + (r - l) / 2;
            if (hull[mid + 1].eval(x) <= hull[mid].eval(x)) l
= mid + 1;
            else r = mid;
        }
        return hull[l].eval(x);
    }
};
```

Data Structure Optimization

```
// DP Optimization with Data Structures (Segment Tree /
Fenwick) - O(N log N)
// Optimiza transiciones: dp[i] = max_{j < i, L <= val[j] <=
R} { dp[j] } + cost[i]

// ─── 1. Compresion de Coordenadas
──────────────────────────────────────
// Mapea valores arbitrarios a [0, sz(vals)-1]
template<class T>
vi compress(const vector<T>& a) {
    vector<T> vals = a;
    sort(all(vals));
    vals.erase(unique(all(vals)), vals.end());
    vi res(sz(a));
    rep(i, sz(a)) {
        res[i] = lower_bound(all(vals), a[i]) - vals.begin();
    }
    return res;
}

// ─── 2. Segment Tree para Maximos en Rango
──────────────────────────────
struct SegTreeDP {
    int n;
    vll tree;
    SegTreeDP(int n) : n(n), tree(4 * n, 0) {}

    void update(int pos, ll val, int node, int l, int r) {
        if (l == r) {
            ckmax(tree[node], val);
            return;
        }
        int mid = l + (r - l) / 2;
        if (pos <= mid) update(pos, val, 2 * node, l, mid);
        else update(pos, val, 2 * node + 1, mid + 1, r);
        tree[node] = max(tree[2 * node], tree[2 * node + 1]);
    }
    void update(int pos, ll val) { update(pos, val, 1, 0, n -
1); }

    ll query(int ql, int qr, int node, int l, int r) {
        if (ql > r || qr < l) return 0; // valor neutro
        if (ql <= l && r <= qr) return tree[node];
        int mid = l + (r - l) / 2;
        return max(query(ql, qr, 2 * node, l, mid),
                   query(ql, qr, 2 * node + 1, mid + 1, r));
    }
    ll query(int ql, int qr) {
        if (ql > qr) return 0;
        return query(ql, qr, 1, 0, n - 1);
    }
};

// ─── 3. Fenwick de Maximos (O(log N) - Solo para Prefijos
[0..idx]) ──────────
struct BITMax {
    int n;
    vll bit;
    BITMax(int n) : n(n), bit(n + 1, 0) {}
```

```
    void update(int i, ll val) {
        for (++i; i <= n; i += i & -i) ckmax(bit[i], val);
    }
    ll query(int i) { // maximo en [0..i]
        ll res = 0;
        for (++i; i > 0; i -= i & -i) ckmax(res, bit[i]);
        return res;
    }
};

// Ejemplo de uso (LIS Ponderado / Flowers):
// vi comp = compress(alturas);
// SegTreeDP st(sz(comp));
// rep(i, n) {
//     ll best_prev = st.query(0, comp[i] - 1);
//     st.update(comp[i], best_prev + bellezas[i]);
// }
// ll ans = st.query(0, sz(comp) - 1);
```

Digit Dp

```
// Digit DP Template - Complejidad O(Longitud * Estados * 10)
// Estados tipicos: (pos, tight, leading_zero,
estado_personalizado)

// ─── 1. Enfoque Estandar f(R) - f(L - 1)
──────────────────────────────────
string S;
ll memo[20][2][2][100]; // Ajustar dimensiones del estado
personalizado

ll digitDP(int pos, bool tight, bool lz, int state) {
    if (pos == sz(S)) {
        // Retornar 1 si el 'state' es valido, 0 si no
        return 1;
    }
    ll& ans = memo[pos][tight][lz][state];
    if (ans != -1) return ans;

    ans = 0;
    int limit = tight ? (S[pos] - '0') : 9;

    for (int dig = 0; dig <= limit; ++dig) {
        bool next_tight = tight && (dig == limit);
        bool next_lz = lz && (dig == 0);

        // Actualizar estado personalizado (ejemplo: suma de
digitos, bitmask, mod, etc.)
        int next_state = next_lz ? 0 : (state + dig);

        ans += digitDP(pos + 1, next_tight, next_lz,
next_state);
        // Si pide modulo: ans %= MOD;
    }
    return ans;
}

ll countValid(ll n) {
    if (n < 0) return 0;
    S = to_string(n);
    memset(memo, -1, sizeof(memo));
    return digitDP(0, true, true, 0);
}

// Rango [L, R]
ll solveRange(ll L, ll R) {
    return countValid(R) - countValid(L - 1);
}

// ─── 2. Enfoque Simultaneo [L, R] (Tight Inferior y
Superior) ─────────────────
// string A, B; // Rellenar A con '0' a la izquierda hasta que
sz(A) == sz(B)
// ll dp2[20][2][2][2]; // pos, tight_low, tight_high,
leading_zero
// int min_d = tight_low  ? (A[pos] - '0') : 0;
// int max_d = tight_high ? (B[pos] - '0') : 9;
// for (int d = min_d; d <= max_d; ++d) {
//     bool n_low  = tight_low  && (d == min_d);
//     bool n_high = tight_high && (d == max_d);
//     bool n_lz   = lz && (d == 0);
//     ...
// }
```

Divide And Conquer

```
// Divide and Conquer DP Optimization - O(K * N log N)
// Condicion: opt[k][i] <= opt[k][i+1] (e.g., Quadrangle
```

```
Inequality)
// Recurrencia: dp[k][i] = min_{j < i} { dp[k-1][j] + cost(j,
i) }

vll dp_prev, dp_curr;

// 1. Costo O(1) con formulas / prefijos (personalizar segun
problema)
ll cost(int j, int i) {
    // Ejemplo: (pref[i] - pref[j])^2
    // ll s = pref[i] - pref[j];
    // return s * s;
    return 0;
}

// 2. Variante: Costo con Ventana Dinamica / Two Pointers
estilo Mo
// int curL = 1, curR = 0; ll cur_cost = 0;
// void move_to(int L, int R) {
//     while (curR < R) add(++curR);
//     while (curL > L) add(--curL);
//     while (curR > R) remove(curR--);
//     while (curL < L) remove(curL++);
// }

void compute(int l, int r, int optL, int optR) {
    if (l > r) return;
    int mid = l + (r - l) / 2;
    int best_opt = -1;
    dp_curr[mid] = LINF;
    int limit = min(mid - 1, optR);

    for (int j = optL; j <= limit; ++j) {
        if (dp_prev[j] == LINF) continue;
        // Si se usa ventana dinamica: move_to(j + 1, mid);
        ll val = dp_prev[j] + cost(j, mid); // o dp_prev[j] +
cur_cost;
        if (val < dp_curr[mid]) {
            dp_curr[mid] = val;
            best_opt = j;
        }
    }

    compute(l, mid - 1, optL, best_opt);
    compute(mid + 1, r, best_opt, optR);
}

// Resuelve para K particiones en un arreglo de tamaño N (1-
indexed)
ll solveDCDP(int n, int k) {
    dp_prev.assign(n + 1, LINF);
    dp_curr.assign(n + 1, LINF);

    // Caso base k = 1
    dp_prev[0] = 0;
    rep1(i, n) dp_prev[i] = cost(0, i);

    // Transicion para k = 2..K
    for (int c = 2; c <= k; ++c) {
        compute(1, n, 0, n - 1);
        dp_prev = dp_curr;
    }
    return dp_prev[n];
}
```

Knapsack

```
// 0/1 Knapsack - O(N * W)
// items: {peso, valor}
ll knapsack(const vii& items, int W) {
    vll dp(W + 1, -1);
    dp[0] = 0;
    each(item, items) {
        int w = item.ff;
        ll v = item.ss;
        for (int j = W; j >= w; --j) {
            if (dp[j - w] != -1) {
                ckmax(dp[j], dp[j - w] + v);
            }
        }
    }
    ll max_v = 0;
    rep(j, W + 1) ckmax(max_v, dp[j]);
    return max_v;
}
```

Knapsack Binary

```
// Bounded Knapsack (Binary Splitting) - O(W * sum(log K))
ll knapsackBounded(const vi& pesos, const vll& valores, const
vi& copias, int W) {
    vii items;
    int n = sz(pesos);
    rep(i, n) {
        int k = copias[i], b = 1;
        while (k > 0) {
            int num = min(k, b);
            items.eb(pesos[i] * num, valores[i] * num);
            k -= num;
            b *= 2;
        }
    }
    vll dp(W + 1, -1);
    dp[0] = 0;
    each(item, items) {
        int w = item.ff;
        ll v = item.ss;
        for (int j = W; j >= w; --j) {
            if (dp[j - w] != -1) {
                ckmax(dp[j], dp[j - w] + v);
            }
        }
    }
    ll ans = 0;
    rep(j, W + 1) ckmax(ans, dp[j]);
    return ans;
}
```

Knapsack Re

```
// 0/1 Knapsack with Reconstruction - O(N * W)
// items: {peso, valor} -> retorna {max_valor,
indices_tomados}
pair<ll, vi> knapsackReconstruction(const vii& items, int W) {
    int n = sz(items);
    vvll dp(n + 1, vll(W + 1, 0));
    rep(i, n) {
        int w = items[i].ff;
        ll v = items[i].ss;
        rep(j, W + 1) {
            dp[i + 1][j] = dp[i][j];
            if (j >= w) ckmax(dp[i + 1][j], dp[i][j - w] + v);
        }
    }
    vi taken;
    int cur_w = W;
    for (int i = n - 1; i >= 0; --i) {
        int w = items[i].ff;
        ll v = items[i].ss;
        if (cur_w >= w && dp[i + 1][cur_w] == dp[i][cur_w - w]
+ v) {
            taken.pb(i);
            cur_w -= w;
        }
    }
    reverse(all(taken));
    return {dp[n][W], taken};
}
```

Knapsack Subset Sum

```
// Subset Sum with Equal Partition / Difference DP - O(N *
sum)
// Encuentra la máxima altura igual de dos subconjuntos
disjuntos (Tallest Billboard)
int knapsackDiferencia(const vi& arr) {
    int sum = 0;
    each(x, arr) sum += x;
    vi dp(sum + 1, -1);
    dp[0] = 0;

    each(h, arr) {
        vi next_dp = dp;
        rep(d, sum + 1) {
            if (dp[d] < 0) continue;
            // 1. No usar h (ya cubierto por next_dp = dp)
            // 2. Añadir al lado mayor
            if (d + h <= sum) ckmax(next_dp[d + h], dp[d]);
            // 3. Añadir al lado menor
            int diff = abs(d - h);
            ckmax(next_dp[diff], dp[d] + min(d, h));
        }
        dp = move(next_dp);
    }
```

```
    return dp[0];
}
```

Knuth

```
// Knuth's DP Optimization - O(N^2)
// Recurrencia: dp[i][j] = min_{i <= k < j} { dp[i][k] +
dp[k+1][j] } + cost(i, j)
// Condicion de optimo: opt[i][j-1] <= opt[i][j] <= opt[i+1]
[j]
// Requiere que cost(i, j) cumpla Quadrangle Inequality y
Monotonicidad

ll knuthDP(const vll& arr) {
    int n = sz(arr);
    // Prefijos para calcular costo de rango en O(1)
    vll pref(n + 1, 0);
    rep(i, n) pref[i + 1] = pref[i] + arr[i];

    auto cost = [&](int i, int j) -> ll {
        return pref[j + 1] - pref[i];
    };

    vvll dp(n, vll(n, 0));
    vvi opt(n, vi(n, 0));

    // Casos base longitud 1
    rep(i, n) {
        dp[i][i] = 0;
        opt[i][i] = i;
    }

    // Iterar por longitud del intervalo len = 2..N
    // O equivalentemente i de n-2 hasta 0, j de i+1 hasta n-1
    for (int len = 2; len <= n; ++len) {
        for (int i = 0; i <= n - len; ++i) {
            int j = i + len - 1;
            dp[i][j] = LINF;
            ll c = cost(i, j);

            int l_opt = opt[i][j - 1];
            int r_opt = (i + 1 <= j) ? opt[i + 1][j] : j - 1;
            r_opt = min(r_opt, j - 1);

            for (int k = l_opt; k <= r_opt; ++k) {
                ll val = dp[i][k] + dp[k + 1][j] + c;
                if (val < dp[i][j]) {
                    dp[i][j] = val;
                    opt[i][j] = k;
                }
            }
        }
    }
    return dp[0][n - 1];
}
```

LCS

```
// Longest Common Subsequence & Reconstruction - O(N * M)
pair<int, string> LCS(const string& s1, const string& s2) {
    int n = sz(s1), m = sz(s2);
    vvi dp(n + 1, vi(m + 1, 0));
    rep(i, n) rep(j, m) {
        if (s1[i] == s2[j]) dp[i + 1][j + 1] = dp[i][j] + 1;
        else dp[i + 1][j + 1] = max(dp[i][j + 1], dp[i + 1]
[j]);
    }
    string ans = "";
    int i = n, j = m;
    while (i > 0 && j > 0) {
        if (s1[i - 1] == s2[j - 1]) {
            ans += s1[i - 1];
            i--; j--;
        } else if (dp[i - 1][j] >= dp[i][j - 1]) {
            i--;
        } else {
            j--;
        }
    }
    reverse(all(ans));
    return {dp[n][m], ans};
}
```

LIS

```
// O(N^2) Longest Increasing Subsequence
int LIS(const vi& arr) {
```

```
    int n = sz(arr);
    vi dp(n, 1);
    int res = 0;
    rep(i, n) {
        rep(j, i) {
            if (arr[j] < arr[i]) ckmax(dp[i], dp[j] + 1);
        }
        ckmax(res, dp[i]);
    }
    return res;
}

vi LISReconstruccion(const vi& arr) {
    int n = sz(arr);
    vi dp(n, 1), padre(n, -1);
    int res = 0, best_idx = -1;

    rep(i, n) {
        rep(j, i) {
            if (arr[j] < arr[i] && dp[j] + 1 > dp[i]) {
                dp[i] = dp[j] + 1;
                padre[i] = j;
            }
        }
        if (dp[i] > res) {
            res = dp[i];
            best_idx = i;
        }
    }

    vi secuencia;
    for (int i = best_idx; i != -1; i = padre[i]) {
        secuencia.pb(arr[i]);
    }
    reverse(all(secuencia));
    return secuencia;
}
```

Lisfast

```
// LIS Fast O(N log N)
int LISFast(const vi& arr) {
    vi res;
    each(x, arr) {
        auto it = lower_bound(all(res), x); // upper_bound
para no estrictamente creciente
        if (it == res.end()) res.pb(x);
        else *it = x;
    }
    return sz(res);
}

// LIS Fast con Reconstrucción - O(N log N)
vi LISFastReconstruccion(const vi& arr) {
    int n = sz(arr);
    vi res, resIndex, padre(n, -1);

    rep(i, n) {
        int x = arr[i];
        auto it = lower_bound(all(res), x);
        int pos = it - res.begin();

        if (it == res.end()) {
            res.pb(x);
            resIndex.pb(i);
        } else {
            *it = x;
            resIndex[pos] = i;
        }

        if (pos > 0) padre[i] = resIndex[pos - 1];
    }

    vi lis;
    int idx = resIndex.back();
    while (idx != -1) {
        lis.pb(arr[idx]);
        idx = padre[idx];
    }
    reverse(all(lis));
    return lis;
}
```

Maximum Subarray

```
// Kadane's Algorithm - Maximum Subarray Sum O(N)
ll maximumSub(const vll& arr) {
    ll cur = 0, ans = -LINF;
    each(x, arr) {
        cur = max(x, cur + x);
        ckmax(ans, cur);
    }
    return ans;
}
```

Subset Sum Bitset

```
// Subset Sum con Bitset & Binary Splitting - O((S / 64) *
sum(log count))
// 1. Bitset acelera transiciones booleanas dividiendo entre
64.
// 2. Binary splitting agrupa elementos repetidos: si 'val'
aparece C veces,
//    se descompone en potencias de 2: 1*val, 2*val, 4*val,
..., R*val.

const int MAX_SUM = 500005;

// ─── 1. Bounded Subset Sum con Bitset Estandar
──────────────────────────────
bitset<MAX_SUM> subsetSum(const vi& arr, int target) {
    // 1. Contar frecuencias
    map<int, int> freq;
    each(x, arr) if (x <= target) freq[x]++;

    // 2. Descomposicion binaria de frecuencias
    vi items;
    for (auto& [val, count] : freq) {
        int c = min(count, target / val);
        for (int k = 1; k <= c; k *= 2) {
            items.pb(val * k);
            c -= k;
        }
        if (c > 0) items.pb(val * c);
    }
    // 3. DP con Bitset
    bitset<MAX_SUM> dp;
    dp[0] = 1;
    each(w, items) {
        dp |= (dp << w);
    }
    return dp;
}

// ─── 2. Bitset con Tamaño Dinamico (Template Recursivo)
──────────────────────
// Ejecuta con bitset<len> ajustado al target para no
desperdiciar operaciones
int reachable_max = 0;
template <int len = 1>
void dynamicSubsetSum(int target, const vi& items) {
    if (target >= len) {
        dynamicSubsetSum<min(len * 2, MAX_SUM)>(target,
items);
        return;
    }
    bitset<len> dp;
    dp[0] = 1;
    each(x, items) dp |= (dp << x);

    for (int i = target; i >= 0; --i) {
        if (dp[i]) {
            reachable_max = i;
            break;
        }
    }
}

// Ejemplo de uso: maxima suma posible <= target con binary
splitting
int getMaxSubsetSum(const vi& valores, int target) {
    map<int, int> freq;
    each(x, valores) if (x <= target) freq[x]++;
    vi items;
    for (auto& [val, count] : freq) {
        int c = min(count, target / val);
        for (int k = 1; k <= c; k *= 2) {
            items.pb(val * k);
            c -= k;
```

```
        }
        if (c > 0) items.pb(val * c);
    }
    reachable_max = 0;
    dynamicSubsetSum(target, items);
    return reachable_max;
}


// ─── 3. Reconstruccion de Elementos Tomados
──────────────────────────────
vi reconstructSubsetSum(const vi& arr, int target) {
    int n = sz(arr);
    // Para reconstruccion exacta se pueden guardar bitsets
por capas o usar backtrack
    vector<bitset<MAX_SUM>> dp(n + 1);
    dp[0][0] = 1;
    rep(i, n) {
        dp[i + 1] = dp[i] | (dp[i] << arr[i]);
    }
    if (!dp[n][target]) return {}; // Imposible

    vi taken;
    int cur = target;
    for (int i = n - 1; i >= 0; --i) {
        if (cur >= arr[i] && dp[i][cur - arr[i]]) {
            taken.pb(arr[i]);
            cur -= arr[i];
        }
    }
    return taken;
}
```

Data Structures

Binary Lifting Weighted

```
struct BinLiftW {
  int n, K;
  vector<vector<pair<int,ll>>> g;
  vector<vector<int>> up;
  vector<vector<ll>> mx; // mx[k][v] = max edge weight on the
2^k jump from v
  vector<int> dep;
  ll merge(ll a, ll b) { return max(a,b); } // your merge here
  const ll ID = LLONG_MIN; // identity of merge
  BinLiftW(int n) : n(n), K(__lg(n)+1), g(n), up(K,vector<int>
(n)), mx(K,vector<ll>(n,LLONG_MIN)), dep(n) {}
  void add_edge(int a, int b, ll w) { g[a].push_back({b,w});
g[b].push_back({a,w}); }
  void build(int root = 0) {
    vector<int> s = {root};
    up[0][root] = root;
    while (!s.empty()) {
      int u = s.back(); s.pop_back();
      for (auto [v,w] : g[u]) if (v != up[0][u]) { up[0][v] =
u; mx[0][v] = w; dep[v] = dep[u]+1; s.push_back(v); }
    }
    for (int k = 1; k < K; k++)
      for (int v = 0; v < n; v++) {
        int m = up[k-1][v];
        up[k][v] = up[k-1][m];
        mx[k][v] = merge(mx[k-1][v],mx[k-1][m]);
      }
  }
  ll query(int a, int b) { // aggregate over edges of the path
a-b
    ll res = ID;
    if (dep[a] < dep[b]) swap(a,b);
    for (int k = 0; k < K; k++)
      if ((dep[a]-dep[b])>>k & 1) { res = merge(res,mx[k][a]);
a = up[k][a]; }
    if (a == b) return res;
    for (int k = K-1; k >= 0; k--)
      if (up[k][a] != up[k][b]) {
        res = merge(res,merge(mx[k][a],mx[k][b]));
        a = up[k][a]; b = up[k][b];
      }
    return merge(res,merge(mx[0][a],mx[0][b]));
  }
};
```

Binary Lifting

```
struct BinLift {
  int n, K;
  vector<vector<int>> g, up;
```

```
  vector<int> dep;
  BinLift(int n) : n(n), K(__lg(n)+1), g(n), up(K,vector<int>
(n)), dep(n) {}
  void add_edge(int a, int b) { g[a].push_back(b);
g[b].push_back(a); }
  void build(int root = 0) {
    vector<int> s = {root};
    up[0][root] = root; // root is its own parent
    while (!s.empty()) { // iterative dfs
      int u = s.back(); s.pop_back();
      for (int v : g[u]) if (v != up[0][u]) { up[0][v] = u;
dep[v] = dep[u]+1; s.push_back(v); }
    }
    for (int k = 1; k < K; k++)
      for (int v = 0; v < n; v++) up[k][v] = up[k-1][up[k-1]
[v]];
  }
  int kth(int u, int k) { // k-th ancestor of u, root if k >
dep[u]
    for (int i = 0; i < K; i++) if (k>>i & 1) u = up[i][u];
    return u;
  }
  int lca(int a, int b) {
    if (dep[a] < dep[b]) swap(a,b);
    a = kth(a,dep[a]-dep[b]);
    if (a == b) return a;
    for (int k = K-1; k >= 0; k--)
      if (up[k][a] != up[k][b]) { a = up[k][a]; b = up[k][b];
}
    return up[0][a];
  }
  int dist(int a, int b) { return
dep[a]+dep[b]-2*dep[lca(a,b)]; }
};
```

DSU Rollback

```
struct DSURollback {
  int n, comps;
  vector<int> p, siz;
  vector<pair<int,int>> st; // history: (child root, parent
root), (-1,-1) if no merge
  DSURollback(int n) : n(n), comps(n), p(n), siz(n,1) {
iota(p.begin(),p.end(),0); }
  int find(int x) { while (p[x] != x) x = p[x]; return x; } //
no path compression
  bool unite(int a, int b) {
    a = find(a); b = find(b);
    if (a == b) { st.push_back({-1,-1}); return false; }
    if (siz[a] < siz[b]) swap(a,b);
    p[b] = a; siz[a] += siz[b]; comps--;
    st.push_back({b,a});
    return true;
  }
  void rollback() { // undoes the last unite (even if it did
not merge)
    auto [b,a] = st.back(); st.pop_back();
    if (b < 0) return;
    p[b] = b; siz[a] -= siz[b]; comps++;
  }
  bool same(int a, int b) { return find(a) == find(b); }
  int size(int x) { return siz[find(x)]; }
};
```

DSU

```
struct DSU {
  int n;
  vector<int> p, siz;
  DSU(int n) : n(n), p(n), siz(n,1) {
iota(p.begin(),p.end(),0); }
  int find(int x) { return p[x] == x ? x : p[x] = find(p[x]);
}
  bool unite(int a, int b) { // false if already in the same
set
    a = find(a); b = find(b);
    if (a == b) return false;
    if (siz[a] < siz[b]) swap(a,b);
    p[b] = a; siz[a] += siz[b];
    return true;
  }
  bool same(int a, int b) { return find(a) == find(b); }
  int size(int x) { return siz[find(x)]; }
};
```

Fenwick Tree 2D Range

```
struct Fenwick2DRange {
  int n, m;
  vector<vector<ll>> b[4];
  Fenwick2DRange(int n, int m) : n(n), m(m) {
    for (auto& b : b) b.assign(n+1,vector<ll>(m+1));
  }
  void add(int x, int y, ll v) { // point update on the
difference array
    ll c[4] = {v, v*x, v*y, v*x*y};
    for (int i = x+1; i <= n; i += i&-i)
      for (int j = y+1; j <= m; j += j&-j)
        for (int k = 0; k < 4; k++) b[k][i][j] += c[k];
  }
  void update(int x1, int y1, int x2, int y2, ll v) { // [x1,
x2) x [y1, y2) += v
    add(x1,y1,v); add(x1,y2,-v); add(x2,y1,-v); add(x2,y2,v);
  }
  ll pre(int x, int y) { // sum of [0, x) x [0, y)
    ll s[4] = {};
    for (int i = x; i > 0; i -= i&-i)
      for (int j = y; j > 0; j -= j&-j)
        for (int k = 0; k < 4; k++) s[k] += b[k][i][j];
    return s[0]*x*y - s[1]*y - s[2]*x + s[3];
  }
  ll query(int x1, int y1, int x2, int y2) { // [x1, x2) x
[y1, y2)
    return pre(x2,y2) - pre(x1,y2) - pre(x2,y1) + pre(x1,y1);
  }
};
```

Fenwick Tree 2D

```
struct Fenwick2D {
  int n, m;
  vector<vector<ll>> t;
  Fenwick2D(int n, int m) : n(n), m(m), t(n+1,vector<ll>(m+1))
{}
  Fenwick2D(const vector<vector<ll>>& a) :
Fenwick2D(a.size(),a[0].size()) {
    for (int i = 0; i < n; i++) for (int j = 0; j < m; j++)
add(i,j,a[i][j]);
  }
  void add(int x, int y, ll v) {
    for (int i = x+1; i <= n; i += i&-i)
      for (int j = y+1; j <= m; j += j&-j) t[i][j] += v;
  }
  ll pre(int x, int y) { // sum of [0, x) x [0, y)
    ll s = 0;
    for (int i = x; i > 0; i -= i&-i)
      for (int j = y; j > 0; j -= j&-j) s += t[i][j];
    return s;
  }
  ll query(int x1, int y1, int x2, int y2) { // [x1, x2) x
[y1, y2)
    return pre(x2,y2) - pre(x1,y2) - pre(x2,y1) + pre(x1,y1);
  }
  void set(int x, int y, ll v) { add(x,y,v -
query(x,y,x+1,y+1)); }
};
```

Fenwick Tree Range

```
struct FenwickRange {
  int n;
  vector<ll> b1, b2;
  FenwickRange(int n) : n(n), b1(n+1), b2(n+1) {}
  void add(vector<ll>& b, int i, ll v) { for (i++; i <= n; i
+= i&-i) b[i] += v; }
  ll sum(vector<ll>& b, int i) { ll s = 0; for (; i > 0; i -=
i&-i) s += b[i]; return s; }
  void update(int l, int r, ll v) { // a[l..r-1] += v
    add(b1,l,v); add(b1,r,-v);
    add(b2,l,v*l); add(b2,r,-v*r);
  }
  ll pre(int i) { return sum(b1,i)*i - sum(b2,i); } // sum of
[0, i)
  ll query(int l, int r) { return pre(r) - pre(l); } // [l, r)
};
```

Fenwick Tree

```
struct Fenwick {
  int n;
  vector<ll> t;
  Fenwick(int n) : n(n), t(n+1) {}
  Fenwick(const vector<ll>& a) : n(a.size()), t(n+1) { // O(n)
```

```
    for (int i = 1; i <= n; i++) {
      t[i] += a[i-1];
      int j = i + (i&-i);
      if (j <= n) t[j] += t[i];
    }
  }
  void add(int i, ll v) { for (i++; i <= n; i += i&-i) t[i] +=
v; }
  ll pre(int i) { // sum of [0, i)
    ll s = 0;
    for (; i > 0; i -= i&-i) s += t[i];
    return s;
  }
  ll query(int l, int r) { return pre(r) - pre(l); } // [l, r)
  void set(int i, ll v) { add(i, v - query(i,i+1)); }
  int lower_bound(ll v) { // first i with pre(i+1) >= v,
(values >= 0), n if none
    int pos = 0;
    for (int k = 1<<__lg(n); k; k >>= 1)
      if (pos+k <= n && t[pos+k] < v) { pos += k; v -= t[pos];
}
    return pos;
  }
};
```

HLD

```
struct HLDNode { ll val = 0; };
struct HLDTree {
  int n;
  vector<HLDNode> t;
  vector<ll> lazy;
  HLDTree(int n) : n(n), t(4*n), lazy(4*n) {}
  HLDNode merge(const HLDNode& a, const HLDNode& b) { return
{a.val + b.val}; }
  void apply(int x, int l, int r, ll v) { t[x].val += v * (r-
l); lazy[x] += v; }
  void push(int x, int l, int r) {
    if (!lazy[x] || r-l == 1) return;
    int m=(l+r)/2; apply(2*x+1,l,m,lazy[x]);
apply(2*x+2,m,r,lazy[x]); lazy[x]=0;
  }
  void build(const vector<ll>& a, int x, int l, int r) {
    if (r-l==1) { t[x].val=a[l]; return; }
    int m=(l+r)/2; build(a,2*x+1,l,m); build(a,2*x+2,m,r);
t[x]=merge(t[2*x+1],t[2*x+2]);
  }
  void build(const vector<ll>& a) { build(a,0,0,n); }
  void update(int ql,int qr,ll v,int x,int l,int r) {
    if (qr<=l || r<=ql) return;
    if (ql<=l && r<=qr) { apply(x,l,r,v); return; }
    push(x,l,r); int m=(l+r)/2; update(ql,qr,v,2*x+1,l,m);
update(ql,qr,v,2*x+2,m,r); t[x]=merge(t[2*x+1],t[2*x+2]);
  }
  void update(int l,int r,ll v) { update(l,r,v,0,0,n); }
  HLDNode query(int ql,int qr,int x,int l,int r) {
    if (ql<=l && r<=qr) return t[x];
    push(x,l,r); int m=(l+r)/2;
    if (qr<=m) return query(ql,qr,2*x+1,l,m);
    if (ql>=m) return query(ql,qr,2*x+2,m,r);
    return
merge(query(ql,qr,2*x+1,l,m),query(ql,qr,2*x+2,m,r));
  }
  HLDNode query(int l,int r) { return query(l,r,0,0,n); }
};

struct HLD {
  int n;
  vector<vector<int>> g;
  vector<int> par, dep, siz, heavy, head, pos;
  HLDTree st;
  HLD(int n) : n(n), g(n), par(n,-1), dep(n), siz(n,1),
heavy(n,-1), head(n), pos(n), st(n) {}
  void add_edge(int a, int b) { g[a].push_back(b);
g[b].push_back(a); }
  void build(const vector<ll>& a, int root = 0) {
    vector<int> ord, s = {root};
    while (!s.empty()) { // preorder, sets parent and depth
      int u = s.back(); s.pop_back(); ord.push_back(u);
      for (int v : g[u]) if (v != par[u]) { par[v] = u; dep[v]
= dep[u]+1; s.push_back(v); }
    }
    for (int i = n-1; i > 0; i--) siz[par[ord[i]]] +=
siz[ord[i]];
    for (int i = n-1; i > 0; i--) {
      int u = ord[i], p = par[u];
      if (heavy[p] < 0 || siz[u] > siz[heavy[p]]) heavy[p] =
u;
```

```
    }
    int cur = 0; head[root] = root; s = {root};
    while (!s.empty()) { // heavy child is pushed last, so it
is visited right after u
      int u = s.back(); s.pop_back(); pos[u] = cur++;
      for (int v : g[u]) if (v != par[u] && v != heavy[u]) {
head[v] = v; s.push_back(v); }
      if (heavy[u] >= 0) { head[heavy[u]] = head[u];
s.push_back(heavy[u]); }
    }
    vector<ll> b(n);
    for (int i = 0; i < n; i++) b[pos[i]] = a[i];
    st.build(b);
  }
  void update_path(int a, int b, ll v) { // vertices on the
path a-b
    for (; head[a] != head[b]; a = par[head[a]]) {
      if (dep[head[a]] < dep[head[b]]) swap(a,b);
      st.update(pos[head[a]],pos[a]+1,v);
    }
    if (dep[a] > dep[b]) swap(a,b);
    st.update(pos[a],pos[b]+1,v);
  }
  HLDNode query_path(int a, int b) {
    HLDNode res; // identity of merge
    for (; head[a] != head[b]; a = par[head[a]]) {
      if (dep[head[a]] < dep[head[b]]) swap(a,b);
      res = st.merge(res,st.query(pos[head[a]],pos[a]+1));
    }
    if (dep[a] > dep[b]) swap(a,b);
    return st.merge(res,st.query(pos[a],pos[b]+1));
  }
  void update_subtree(int u, ll v) {
st.update(pos[u],pos[u]+siz[u],v); }
  HLDNode query_subtree(int u) { return
st.query(pos[u],pos[u]+siz[u]); }
  int lca(int a, int b) {
    for (; head[a] != head[b]; a = par[head[a]])
      if (dep[head[a]] < dep[head[b]]) swap(a,b);
    return dep[a] < dep[b] ? a : b;
  }
};

// Ejemplo de uso anterior (main comentado para compilar con
la plantilla):
// int main() {
//   int n = 5;
//   HLD h(n);
//   h.add_edge(0,1); h.add_edge(0,2); h.add_edge(1,3);
h.add_edge(1,4);
//   h.build({1,2,3,4,5});
//   h.update_path(3,2,10);
//   cout << h.query_path(4,2).val << "\n";
//   cout << h.query_subtree(1).val << "\n";
// }
```

Ordered Set

```
#include <bits/stdc++.h>
#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>
using namespace std;
using namespace __gnu_pbds;

template <class T>
using ordered_set = tree<
T,
  null_type,
  less<T>,                // use less_equal<T> if you want to
allow duplicates (see note below)
  rb_tree_tag,
  tree_order_statistics_node_update>;

// --- Example usage ---

// Ejemplo de uso anterior (comentado para pegar el algoritmo
con la plantilla):
// int main() {
//   ios::sync_with_stdio(false);
//   cin.tie(nullptr);
//
//   ordered_set<int> s;
//
//   s.insert(10);
//   s.insert(20);
//   s.insert(30);
//   s.insert(40);
//
```

```
//   // [1] Get element by index (0-based)
//   cout << "Element at index 2: " << *s.find_by_order(2) <<
"\n"; // -> 30
//
//   // [2] Get index of an element
//   cout << "Index of 30: " << s.order_of_key(30) << "\n"; //
-> 2
//
//   // [3] Lower bound behavior
//   cout << "Index where 25 would be inserted: " <<
s.order_of_key(25) << "\n"; // -> 2
//
//   // [4] Iterate through all elements
//   for (auto x : s) cout << x << " ";
//   cout << "\n";
//
//   // [5] Check if an element exists
//   if (s.find(20) != s.end())
//     cout << "20 exists in set\n";
// }
```

Persistent Segment Tree

```
struct Node {
    ll val = 0;
    int lc = 0, rc = 0; // children, 0 = null node
};
struct PerSegTree {
  int n;
  vector<Node> t;
  vector<int> root; // root[i] = root node of version i
  PerSegTree(int n, int cap = 0) : n(n), t(1), root(1,0) {
t.reserve(cap+1); }
  Node merge(const Node& a,const Node& b) {
    Node res;
    res.val = a.val + b.val; // your merge here
    return res;
  }
  int build(const vector<ll>& a, int l, int r) {
    int x = t.size(); t.emplace_back();
    if (r - l == 1) {t[x].val = a[l]; return x; }
    int m = (l+r)/2;
    int lc = build(a,l,m), rc = build(a,m,r);
    t[x] = merge(t[lc],t[rc]); t[x].lc = lc; t[x].rc = rc;
    return x;
  }
  int update(int p, int i, ll v, int l, int r) { // returns
the new node
    int x = t.size(); t.push_back(t[p]);
    if (r - l == 1) {t[x].val += v; return x; } // your apply
here (= v to assign)
    int m = (l+r)/2;
    if (i < m) {int c = update(t[p].lc,i,v,l,m); t[x].lc = c;
}
    else       {int c = update(t[p].rc,i,v,m,r); t[x].rc = c;
}
    int lc = t[x].lc, rc = t[x].rc;
    t[x] = merge(t[lc],t[rc]); t[x].lc = lc; t[x].rc = rc;
    return x;
  }
  Node query(int ql, int qr, int x, int l, int r) {
    if (ql <= l && r <= qr) return t[x];
    int m = (l + r) / 2;
    if (qr <= m) return query(ql,qr,t[x].lc,l,m);
    if (ql >= m) return query(ql,qr,t[x].rc,m,r);
    return merge(query(ql,qr,t[x].lc,l,m),
                 query(ql,qr,t[x].rc,m,r));
  }
  int kth(int a, int b, ll k) { // k-th smallest (0 idx) in
the count difference of roots b and a
    int l = 0, r = n;
    while (r - l > 1) {
      int m = (l+r)/2;
      ll c = t[t[b].lc].val - t[t[a].lc].val;
      if (k < c) {a = t[a].lc; b = t[b].lc; r = m; }
      else       {k -= c; a = t[a].rc; b = t[b].rc; l = m; }
    }
    return l;
  }
  void build(const vector<ll>& a){ root[0] = build(a,0,n); }
  int update(int ver, int i, ll v){
root.push_back(update(root[ver],i,v,0,n)); return
root.size()-1; } // new version id
  Node query(int ver, int l, int r){ return
query(l,r,root[ver],0,n); }
};

// Ejemplo de uso (main comentado para compilar con la
```

```
plantilla):
// int main() {
//   vector<ll> values = {2, 1, 3, 4};
//   PerSegTree pst(values.size());
//   pst.build(values);                  // version 0
//   int version1 = pst.update(0, 1, 5); // new version, index
1 += 5
//   cout << pst.query(0, 0, 4).val << "\n";
//   cout << pst.query(version1, 0, 4).val << "\n";
// }
```

Segment Tree Lazy

```
#include <bits/stdc++.h>
using namespace std;
using ll = long long;
struct Node {
    ll val = 0;
};
struct SegTree {
  int n;
  vector<Node> t;
  vector<ll> lz;
  SegTree(int n) : n(n), t(4*n), lz(4*n) {}
  Node merge(const Node& a,const Node& b) {
    Node res;
    res.val = a.val + b.val;
    return res;
  }
  void apply(int x, int l, int r, ll v) {
    t[x].val += v*(r-l);
    lz[x] += v;
  }
  void push(int x, int l, int r) {
    if (!lz[x]) return;
    int m = (l+r)/2;
    apply(2*x+1,l,m,lz[x]);
    apply(2*x+2,m,r,lz[x]);
    lz[x] = 0;
  }
  void build(const vector<ll>& a, int x, int l, int r) {
    if (r - l == 1) {t[x].val = a[l]; return; }
    int m = (l+r)/2;
    build(a,2*x+1,l,m);
    build(a,2*x+2,m,r);
    t[x]=merge(t[2*x+1],t[2*x+2]);
  }
  void update(int ql, int qr, ll v, int x, int l, int r) {
    if (qr <= l || r <= ql) return;
    if (ql <= l && r <= qr) {apply(x,l,r,v); return; }
    push(x,l,r);
    int m = (l+r)/2;
    update(ql,qr,v,2*x+1,l,m);
    update(ql,qr,v,2*x+2,m,r);
    t[x]=merge(t[2*x+1],t[2*x+2]);
  }
  Node query(int ql, int qr, int x, int l, int r) {
    if (ql <= l && r <= qr) return t[x];
    push(x,l,r);
    int m = (l + r) / 2;
    if (qr <= m) return query(ql,qr,2*x+1,l,m);
    if (ql >= m) return query(ql,qr,2*x+2,m,r);
    return merge(query(ql,qr,2*x+1,l,m),
                 query(ql,qr,2*x+2,m,r));
  }
  void build(const vector<ll>& a){ build(a,0,0,n); }
  void update(int l, int r, ll v){ update(l,r,v,0,0,n); }
  Node query(int l, int r){ return query(l,r,0,0,n); }
};
```

Segment Tree

```
#include <bits/stdc++.h>
using namespace std;
using ll = long long;
struct Node {
    ll val = 0;
};
struct SegTree {
  int n;
  vector<Node> t;
  SegTree(int n) : n(n), t(4*n) {}
  Node merge(const Node& a,const Node& b) {
    Node res;
    return res;
  }
  void update(int i, ll v, int x, int l, int r) {
    if (r - l == 1) {t[x].val = v; return; }
```

```
    int m = (l+r)/2;
    if (i < m)update(i,v,2*x+1,l,m);
    else      update(i,v,2*x+2,m,r);
    t[x]=merge(t[2*x+1],t[2*x+2]);
  }
  Node query(int ql, int qr, int x, int l, int r) {
    if (ql <= l && r <= qr) return t[x];
    int m = (l + r) / 2;
    if (qr <= m) return query(ql,qr,2*x+1,l,m);
    if (ql >= m) return query(ql,qr,2*x+2,m,r);
    return merge(query(ql,qr,2*x+1,l,m),
                 query(ql,qr,2*x+2,m,r));
  }
  void update(int i, ll v){ update(i,v, 0,0,n); }
  Node query(int l, int r){ return query(l,r,0,0,n); }
};
```

Sparse Table

```
struct SparseTable {
  int n;
  vector<vector<ll>> t;
  ll merge(ll a, ll b) {
    return min(a,b);
  }
  SparseTable(const vector<ll>& a) : n(a.size()) {
    int K = __lg(n) + 1;
    t.assign(K, vector<ll>(n));
    t[0] = a;
    for (int k=1; k<K; k++)
      for (int i = 0; i+(1<<k) <= n; i++)
        t[k][i] = merge(t[k-1][i], t[k-1][i+(1<<(k-1))]);
  }
  ll query(int l, int r) { // [l, r), ops (min, max, gcd, and,
or)
    int k = __lg(r-l);
    return merge(t[k][l], t[k][r-(1<<k)]);
  }
  ll query_log(int l, int r) { // [l, r), any associative op
    ll res = 1e18; // identity (0 for sum/xor)
    for (int k = __lg(r-l); k >= 0; k--)
      if (l + (1<<k) <= r) { res = merge(res, t[k][l]); l +=
1<<k; }
    return res;
  }
};
```

Two Pointers

```
// Two Pointers & Sliding Window Aggregation (SWAG)

// 1. Sliding Window Estandar - O(N)
// Ejemplo: Maxima longitud de subarray con condicion monotona
int slidingWindow(const vi& a, int k) {
    int n = sz(a), ans = 0, l = 0;
    // Estado de la ventana (ej. frecuencias, suma)
    // vi freq(MAX_VAL, 0); int distinct = 0;

    rep(r, n) {
        // add(a[r]);
        // while (!valid()) { remove(a[l++]); }
        ckmax(ans, r - l + 1);
    }
    return ans;
}

// 2. SWAG (Sliding Window Aggregation) - O(1) amortizado por
push/pop/query
// Permite operaciones asociativas NO reversibles (ej: min,
max, gcd, matrix mul)
template<class T>
struct SWAG {
    struct Node {
        T val, agg;
    };
    stack<Node> s1, s2;

    T op(T a, T b) { return min(a, b); } // Cambiar operacion
(gcd, max, +, etc.)

    bool empty() { return s1.empty() && s2.empty(); }
    int size() { return sz(s1) + sz(s2); }

    void push(T val) {
        T agg = s2.empty() ? val : op(s2.top().agg, val);
        s2.push({val, agg});
    }
```

```
    void pop() {
        if (s1.empty()) {
            while (!s2.empty()) {
                T val = s2.top().val;
                s2.pop();
                T agg = s1.empty() ? val : op(val,
s1.top().agg);
                s1.push({val, agg});
            }
        }
        s1.pop();
    }

    T query() {
        if (empty()) return 0; // valor neutro
        if (s1.empty()) return s2.top().agg;
        if (s2.empty()) return s1.top().agg;
        return op(s1.top().agg, s2.top().agg);
    }
};
```

Geometry

Convex Hull

```
#include <bits/stdc++.h>

using namespace std;

struct point {
  double x, y;
  point() { x = y = 0; }
  point(double _x, double _y) : x(_x), y(_y) {}
  bool operator == (point const& t) const {
    return x == t.x && y == t.y;
  }
};

int orientation(point a, point b, point c) {
  double v = a.x*(b.y-c.y)+b.x*(c.y-a.y)+c.x*(a.y-b.y);
  if (v < 0) return -1; // clockwise
  if (v > 0) return +1; // counter-clockwise
  return 0;
}

bool cw(point a, point b, point c, bool include_collinear) {
  int o = orientation(a, b, c);
  return o < 0 || (include_collinear && o == 0);
}
bool collinear(point a, point b, point c) { return
orientation(a, b, c) == 0; }

void convex_hull(vector<point>& a, bool include_collinear =
false) {
  point p0 = *min_element(a.begin(), a.end(), [](point a,
point b) {
      return make_pair(a.y, a.x) < make_pair(b.y, b.x);
      });
  sort(a.begin(), a.end(), [&p0](const point& a, const point&
b) {
      int o = orientation(p0, a, b);
      if (o == 0)
      return (p0.x-a.x)*(p0.x-a.x) + (p0.y-a.y)*(p0.y-a.y)
      < (p0.x-b.x)*(p0.x-b.x) + (p0.y-b.y)*(p0.y-b.y);
      return o < 0;
      });
  if (include_collinear) {
    int i = (int)a.size()-1;
    while (i >= 0 && collinear(p0, a[i], a.back())) i--;
    reverse(a.begin()+i+1, a.end());
  }

  vector<point> st;
  for (int i = 0; i < (int)a.size(); i++) {
    while (st.size() > 1 && !cw(st[st.size()-2], st.back(),
a[i], include_collinear))
      st.pop_back();
    st.push_back(a[i]);
  }

  if (include_collinear == false && st.size() == 2 && st[0] ==
st[1])
    st.pop_back();

  a = st;
}
```

```
double area(const vector<point>& fig) {
  double res = 0;
  for (unsigned i = 0; i < fig.size(); i++) {
    point p = i ? fig[i - 1] : fig.back();
    point q = fig[i];
    res += (p.x - q.x) * (p.y + q.y);
  }
  return fabs(res) / 2;
}


// example use case

// Ejemplo de uso anterior (comentado para pegar el algoritmo
con la plantilla):
// void solve() {
//   double n,r;
//   cin >> n >> r;
//
//   vector<point> coords(n);
//   for(int i = 0; i < n; i++) {
//     cin >> coords[i].x >> coords[i].y;
//   }
//
//   auto dist = [](point &a, point&b) {
//     return sqrt(pow(a.x-b.x,2)+pow(a.y-b.y,2));
//   };
//
//   const double PI = 3.141516;
//   double cir = 2 * PI * r;
//   convex_hull(coords);
//
//   double per = 0;
//   coords.push_back(coords[0]);
//   for(int i = 1; i < coords.size(); i++) {
//     per +=  dist(coords[i-1],coords[i]);
//   }
//
//   int ans = round(per+cir);
//   cout << ans << endl;
// }
//
// int main() {
//   ios::sync_with_stdio(false);
//   cin.tie(0);
//   int t = 1;
//   //cin >> t;
//   while(t--)
//     solve();
// }
```

Graphs

2-SAT

```
#include <bits/stdc++.h>

using namespace std;

#define fst first
#define snd second
#define ll long long
#define endl '\n'

const int INF = 1 << 30;
const int MAXN = 1e5+5;

struct TwoSatSolver {
  int n_vars;
  int n_vertices;
  vector<vector<int>> adj, adj_t;
  vector<bool> used;
  vector<int> order, comp;
  vector<bool> assignment;

  TwoSatSolver(int _n_vars) : n_vars(_n_vars), n_vertices(2 *
n_vars), adj(n_vertices), adj_t(n_vertices), used(n_vertices),
order(), comp(n_vertices, -1), assignment(n_vars) {
    order.reserve(n_vertices);
  }
  void dfs1(int v) {
    used[v] = true;
    for (int u : adj[v]) {
      if (!used[u])
        dfs1(u);
    }
```

```
    order.push_back(v);
  }

  void dfs2(int v, int cl) {
    comp[v] = cl;
    for (int u : adj_t[v]) {
      if (comp[u] == -1)
        dfs2(u, cl);
    }
  }

  bool solve_2SAT() {
    order.clear();
    used.assign(n_vertices, false);
    for (int i = 0; i < n_vertices; ++i) {
      if (!used[i])
        dfs1(i);
    }

    comp.assign(n_vertices, -1);
    for (int i = 0, j = 0; i < n_vertices; ++i) {
      int v = order[n_vertices - i - 1];
      if (comp[v] == -1)
        dfs2(v, j++);
    }

    assignment.assign(n_vars, false);
    for (int i = 0; i < n_vertices; i += 2) {
      if (comp[i] == comp[i + 1])
        return false;
      assignment[i / 2] = comp[i] > comp[i + 1];
    }
    return true;
  }

  void add_disjunction(int a, bool na, int b, bool nb) {
    // na and nb signify whether a and b are to be negated
    a = 2 * a ^ na;
    b = 2 * b ^ nb;
    int neg_a = a ^ 1;
    int neg_b = b ^ 1;
    adj[neg_a].push_back(b);
    adj[neg_b].push_back(a);
    adj_t[b].push_back(neg_a);
    adj_t[a].push_back(neg_b);
  }

};

// Ejemplo de uso anterior (comentado para pegar el algoritmo
con la plantilla):
// void solve() {
//   int n,m;
//   cin >> n >> m;
//
//   TwoSatSolver solver(m);
//   for(int i = 0; i < n; i++) {
//     int topping1, topping2;
//     char op1, op2;
//     cin >> op1 >> topping1 >> op2 >> topping2;
//     bool no_good1 = op1 == '-';
//     bool no_good2 = op2 == '-';
//     topping1--,topping2--;
//     solver.add_disjunction(topping1,no_good1, topping2,
no_good2);
//   }
//
//   bool poss = solver.solve_2SAT();
//   if(!poss) {
//     cout << "IMPOSSIBLE" << endl;
//     return;
//   }
//
//   for(int i = 0; i < m; i++) {
//     if(solver.assignment[i]) {
//       cout << "+ ";
//     } else {
//       cout << "- ";
//     }
//   }
// }
//
// int main() {
//   ios::sync_with_stdio(false);
//   cin.tie(nullptr);
//   int t = 1;
//   //cin >> t;
//   while(t--)
```

```
//     solve();
// }
```

Bellman Ford

```
#include <bits/stdc++.h>

using namespace std;

#define fst first
#define snd second
#define ll long long

const ll INF = LONG_LONG_MAX;
const int MAXN = 1e5+5;

struct Edge {
  int u, v;
  ll weight;
};

int n,m;
vector<Edge> edges;

// Bellman-Ford algorithm to detect negative cycles and find
shortest paths
bool bellman(int src, vector<ll> &dist) {
  dist.assign(n + 1, INF);
  dist[src] = 0;
  for(int i = 2; i <= n; i++)
  {
    for (int j = 0; j < m; j++)
    {
      int u = edges[j].u;
      int v = edges[j].v;
      ll weight = edges[j].weight;
      if (dist[u]!=INF && dist[u] + weight < dist[v])
        dist[v] = dist[u] + weight;
    }
  }
  for (int i = 0; i < m; i++)
  {
    int u = edges[i].u;
    int v = edges[i].v;
    ll weight = edges[i].weight;
    // True if neg-cylce exists
    if (dist[u]!=INF && dist[u] + weight < dist[v])
      return true;
  }
  return false;
}

// Ejemplo de uso anterior (comentado para pegar el algoritmo
con la plantilla):
// int main() {
//   cin >> n >> m;
//   edges.resize(m);
//
//   for (int i = 0; i < m; i++) {
//     int a, b;
//     ll w;
//     cin >> a >> b >> w;
//     edges[i] = {a, b, w};
//   }
//
//   vector<ll> dist;
//
//   bool hasNegativeCycle = bellman(1, dist);
//
//   if (hasNegativeCycle) {
//     cout << "negative-weight cycle\n";
//   } else {
//     if (dist[n] == INF) {
//       cout << "no path from 1 to " << n << "\n";
//     } else {
//       cout << "shortest path from 1 to " << n << " is: " <<
dist[n] << "\n";
//     }
//   }
// }
```

Cd

```
#include "bits/stdc++.h"
using namespace std;

using ll = long long;
using ld = long double;
```

```
#define sz(v) ((int)((v).size()))

const int MAXN = 100005;

int n;
vector<int> adj[MAXN];
int sub[MAXN];
bool removed[MAXN];
int parCentroid[MAXN];
vector<vector<pair<int,int>>> elements(MAXN);

vector<vector<pair<int,int>>> sufix(MAXN);
#define linea() cerr << "--------------------------" << '\n'
int dfs_size(int u, int p) {
    sub[u] = 1;
    for (int v : adj[u]) {
        if (v == p || removed[v]) continue;
        sub[u] += dfs_size(v, u);
    }
    return sub[u];
}

int find_centroid(int u, int p, int total) {
    for (int v : adj[u]) {
        if (v == p || removed[v]) continue;
        if (sub[v] > total / 2)
            return find_centroid(v, u, total);
    }
    return u;
}
void bfs(int node) {
    map<int, int> dist;
    queue<int> q;
    q.push(node);
    elements[node].push_back({node, 0});
    while (!q.empty()) {
        auto l = q.front();
        q.pop();
        for (auto& x : adj[l]) {
            if (dist[x] == 0 && !removed[x] && x != node) {
                dist[x] = dist[l] + 1;
                q.push(x);
                elements[node].push_back({x, dist[x]});
            }
        }
    }

    sort(elements[node].begin(), elements[node].end());

    sufix[node].resize(sz(elements[node]) + 1);

    pair<int, int> otro {1e9, 1e9};
    for (int i = sz(elements[node]) - 1; i >= 0; i--) {
        if (otro.second > elements[node][i].second) {
            otro = elements[node][i];
        }
        else if (otro.second == elements[node][i].second) {
            if (otro.first > elements[node][i].first) otro =
elements[node][i];
        }
        sufix[node][i] = otro;
    }
}
void decompose(int u, int p = -1) {
    int total = dfs_size(u, -1);
    int c = find_centroid(u, -1, total);
    bfs(c);
    removed[c] = true;

    parCentroid[c] = (p == -1 ? c : p); // root centroid
points to itself

    // Example: hook custom logic here (before recursing)
    // e.g., preprocess distances from c to other nodes in its
component

    for (int v : adj[c]) {
        if (!removed[v])
            decompose(v, c);
    }
}
int bs(vector<pair<int, int>>& v, int target) {
    int low = 0, high = sz(v) - 1;
    int ans = -1;
    while (low <= high) {
        int mid = low + (high - low) / 2;
        if (v[mid].first > target) {
            ans = mid;
```

```
            high = mid - 1;
        }
        else low = mid + 1;
    }
    return ans;
}
int dist(vector<pair<int, int>>& v, int target) {
    int low = 0, high = sz(v) - 1;
    int ans = -1;
    while (low <= high) {
        int mid = low + (high - low) / 2;
        if (v[mid].first == target) {
            return mid;
        }
        else if (v[mid].first < target) low = mid + 1;
        else high = mid - 1;
    }
    return -1;
}

// Ejemplo de uso anterior (comentado para compilar con la
plantilla):
// int32_t main() {
//     ios::sync_with_stdio(false);
//     cin.tie(nullptr);
//     cin >> n;
//     for (int i = 0; i < n - 1; i++) {
//         int u, v;
//         cin >> u >> v;
//         adj[u].push_back(v);
//         adj[v].push_back(u);
//     }
//
//     decompose(1);
//     for (int i = 1; i < n; i++) {
//         int aux = i;
//         int ori = aux;
//         pair<int, int> res = {1e9, 1e9};
//         bool ya = false;
//         do {
//             int it = bs(elements[aux], ori);
//             if (it == -1) {
//                 aux = parCentroid[aux];
//                 linea();
//                 continue;
//             }
//             pair<int, int> ans = sufix[aux][it];
//             int it2 = dist(elements[aux], ori);
//             ans.second += elements[aux][it2].second;
//             if (ans.second < res.second) {
//                 res = ans;
//             }
//             else if (ans.second == res.second) {
//                 if (ans.first < res.first) {
//                     res = ans;
//                 }
//             }
//             aux = parCentroid[aux];
//             linea();
//             if (ya) break;
//             if (parCentroid[aux] == aux) ya = true;
//         } while (true);
//         cout << res.first << ' ';
//     }
//     cout << n << '\n';
//
//     return 0;
// }
```

Centroid Decomposition

```
#include <bits/stdc++.h>
using namespace std;

#define endl '\n'
using ll = long long;

const int MAXN = 200005;

// ---------- GLOBALS ----------
int n;
vector<int> adj[MAXN];
int sub[MAXN];         // subtree sizes
bool removed[MAXN];    // centroid removed flag
int parCentroid[MAXN]; // parent in centroid tree

int dfs_size(int u, int p) {
    sub[u] = 1;
```

```
    for (int v : adj[u]) {
        if (v == p || removed[v]) continue;
        sub[u] += dfs_size(v, u);
    }
    return sub[u];
}

int find_centroid(int u, int p, int total) {
    for (int v : adj[u]) {
        if (v == p || removed[v]) continue;
        if (sub[v] > total / 2)
            return find_centroid(v, u, total);
    }
    return u;
}

void decompose(int u, int p = -1) {
    int total = dfs_size(u, -1);
    int c = find_centroid(u, -1, total);
    removed[c] = true;

    parCentroid[c] = (p == -1 ? c : p); // root centroid
points to itself

    // Example: hook custom logic here (before recursing)
    // e.g., preprocess distances from c to other nodes in its
component

    for (int v : adj[c]) {
        if (!removed[v])
            decompose(v, c);
    }
}

// ---------- MAIN ----------

// Ejemplo de uso anterior (comentado para pegar el algoritmo
con la plantilla):
// int main() {
//     cin >> n;
//     for (int i = 0; i < n - 1; i++) {
//         int u, v;
//         cin >> u >> v;
//         adj[u].push_back(v);
//         adj[v].push_back(u);
//     }
//
//     decompose(0);
//     cout << "Centroid decomposition parent array:\n";
//     for (int i = 0; i < n; i++)
//         cout << i << " -> " << parCentroid[i] << endl;
//
//     return 0;
// }
```

Dijkstra

```
const ll INF = LLONG_MAX / 4;
struct Dijkstra {
  int n;
  vector<vector<pair<int,ll>>> g;
  vector<ll> d;
  vector<int> par;
  Dijkstra(int n) : n(n), g(n) {}
  void add_edge(int a, int b, ll w) { g[a].push_back({b,w}); }
// directed
  void add_undirected(int a, int b, ll w) { add_edge(a,b,w);
add_edge(b,a,w); }
  void run(const vector<int>& src) { // multi-source
    d.assign(n,INF); par.assign(n,-1);

priority_queue<pair<ll,int>,vector<pair<ll,int>>,greater<>>
pq;
    for (int s : src) { d[s] = 0; pq.push({0,s}); }
    while (!pq.empty()) {
      auto [du,u] = pq.top(); pq.pop();
      if (du > d[u]) continue; // stale entry
      for (auto [v,w] : g[u])
        if (d[u]+w < d[v]) { d[v] = d[u]+w; par[v] = u;
pq.push({d[v],v}); }
    }
  }
  void run(int s) { run(vector<int>{s}); }
  vector<int> path(int t) { // s -> t, empty if unreachable
    vector<int> p;
    if (d[t] >= INF) return p;
    for (; t >= 0; t = par[t]) p.push_back(t);
    reverse(p.begin(),p.end());
```

```
    return p;
  }
};
```

Floyd Warshall

```
#include <bits/stdc++.h>

using namespace std;

#define fst first
#define snd second
#define ll long long

const int INF = 1 << 30;
const int MAXN = 1e3+5;

int n;
ll d[MAXN][MAXN];

void floyd_warshall() {
  for (int k = 1; k <= n; k++)
    for (int i = 1; i <= n; i++)
      for (int j = 1; j <= n; j++)
        if (d[i][k] + d[k][j] < d[i][j])
          d[i][j] = d[i][k] + d[k][j];
}

// Ejemplo de uso anterior (comentado para pegar el algoritmo
con la plantilla):
// void solve() {
//   int s, t;
//   cin >> n;
//   cin >> s >> t;
//   for(int i = 1; i <= n; i++) {
//     for(int j = 1; j <= n; j++) {
//       cin >> d[i][j];
//       if(d[i][j] == -1) d[i][j] = INF;
//     }
//   }
//
//   if(d[s][t] == INF)
//     cout << "-1\n";
//   else
//     cout << d[s][t] << '\n';
// }
//
// int main() {
//   ios::sync_with_stdio(false);
//   cin.tie(nullptr);
//   int t = 1;
//   //cin >> t;
//   while(t--)
//     solve();
// }
```

Ford Fulkerson

```
#include <bits/stdc++.h>

using namespace std;

#define ll long long
#define endl '\n'

const ll INF = 1e18;

struct Dinic {
  struct Edge {
    int to, rev;
    ll cap;
  };

  int n;
  vector<vector<Edge>> g;
  vector<int> level, it;

  Dinic(int n) : n(n), g(n), level(n), it(n) {}

  void add_edge(int u, int v, ll cap) {
    g[u].push_back({v, (int)g[v].size(), cap});
    g[v].push_back({u, (int)g[u].size() - 1, 0});
  }

  bool bfs(int s, int t) {
    fill(level.begin(), level.end(), -1);
    queue<int> q;
    level[s] = 0; q.push(s);
```

```
  while (!q.empty()) {
    int u = q.front(); q.pop();
    for (auto &e : g[u]) {
      if (e.cap > 0 && level[e.to] == -1) {
        level[e.to] = level[u] + 1;
        q.push(e.to);
      }
    }
  }
  return level[t] != -1;
}

ll dfs(int u, int t, ll f) {
  if (u == t) return f;
  for (int &i = it[u]; i < (int)g[u].size(); i++) {
    Edge &e = g[u][i];
    if (e.cap > 0 && level[e.to] == level[u] + 1) {
      ll ret = dfs(e.to, t, min(f, e.cap));
      if (ret > 0) {
        e.cap -= ret;
        g[e.to][e.rev].cap += ret;
        return ret;
      }
    }
  }
  return 0;
}

ll max_flow(int s, int t) {
  ll flow = 0, f;
  while (bfs(s, t)) {
    fill(it.begin(), it.end(), 0);
    while ((f = dfs(s, t, INF)) > 0)
      flow += f;
  }
  return flow;
}

// Min-cut edges
void print_mincut_edges(int s) {
  vector<bool> vis(n, false);
  queue<int> q;
  q.push(s);
  vis[s] = true;
    while (!q.empty()) {
      int u = q.front(); q.pop();
      for (auto &e : g[u]) {
        if (e.cap > 0 && !vis[e.to]) {
          vis[e.to] = true;
          q.push(e.to);
        }
      }
    }
    for (int u = 0; u < n; ++u) {
      if (vis[u]) {
        for (auto &e : g[u]) {
          if (!vis[e.to] && e.cap == 0) {
            cout << u + 1 << " " << e.to + 1 << endl;
          }
        }
      }
    }
  }
};

// Ejemplo de uso anterior (comentado para pegar el algoritmo
con la plantilla):
// int main() {
//   int n,m;
//   cin >> n >> m;
//   Dinic g(n);
//   for (int i = 0; i < m; ++i) {
//     int u, v, w;
//     cin >> u >> v;
//     w = 1;
//     u--;v--;
//     g.add_edge(u, v, w);
//   }
//   cout << g.max_flow(0,n-1) << endl;
//   g.print_mincut_edges(0);
// }
```

Functional Graph

```
#include <bits/stdc++.h>

using namespace std;
```

```
#define endl '\n'

const int MAXN = 2e5+5;
const int LOG = 22;

int up[MAXN][LOG];
bool visited[MAXN];
bool on_stack[MAXN];
int next_node[MAXN];
int cycle[MAXN];
int cycle_length[MAXN];
int depth[MAXN];

int cycle_idx[MAXN];

void mark_cycle(int u, int id) {
  cycle[u] = id;
  depth[u] = 0;
  cycle_length[id]++;
  cycle_idx[u] = cycle_length[id];
  if(!cycle[next_node[u]])
    mark_cycle(next_node[u], id);
}

void dfs(int u) {
  visited[u] = 1;
  on_stack[u] = 1;

  int v = next_node[u];
  if(on_stack[v]) {
    mark_cycle(v, v);
  } else if(!visited[v]){
    dfs(v);
  }

  if(!cycle[u])
    depth[u] = depth[v] + 1;

  on_stack[u] = 0;
}

int jumpto(int u, int k) {
  int v = u;
  for(int i = 0; i < LOG; i++) {
    if((1 << i) & k) {
      v = up[v][i];
    }
  }
  return v;
}


int distcycle(int u, int v) {
  if(cycle_idx[u] < cycle_idx[v]) {
    return cycle_idx[v] - cycle_idx[u];
  }
  int turn = (cycle_length[cycle[u]] - cycle_idx[u] + 1) %
cycle_length[cycle[u]];
  return turn + cycle_idx[v] - 1;
}

// Ejemplo de uso anterior (comentado para pegar el algoritmo
con la plantilla):
// void solve() {
//   int n,q;
//   cin >> n >> q;
//
//   for(int i = 1; i <= n; i++) {
//     cin >> next_node[i];
//     up[i][0] = next_node[i];
//   }
//
//   for(int j = 1; j < LOG; j++) {
//     for(int i = 1; i <= n; i++) {
//       up[i][j] = up[up[i][j-1]][j-1];
//     }
//   }
//
//   for(int i = 1; i <= n; i++) {
//     if(!visited[i]) dfs(i);
//   }
//
//   while(q--) {
//     int u,v;
//     cin >> u >> v;
//
//     int cyu = cycle[u];
//     int cyv = cycle[v];
```

```
//
//     if(cyu == 0) {
//       cyu = cycle[jumpto(u,depth[u])];
//     }
//     if(cyv == 0) {
//       cyv = cycle[jumpto(v,depth[v])];
//     }
//
//     int a = u;
//     int b = v;
//
//     if(cyu != cyv) {
//       cout << -1 << endl;
//       continue;
//     }
//
//     if(cycle[u] == 0 && cycle[v] == 0) {
//       if(depth[u] < depth[v]) {
//         cout << -1 << endl;
//         continue;
//       }
//       int diff = depth[u] - depth[v];
//       if(jumpto(u,diff) != v) {
//         cout << -1 << endl;
//       } else {
//         cout << diff << endl;
//       }
//       continue;
//     }
//
//     if(cycle[u] && cycle[v]) {
//       cout << distcycle(u,v) << endl;
//       continue;
//     }
//
//     if(cycle[u] == 0) {
//       int steps = depth[u];
//       steps += distcycle(jumpto(u,depth[u]), v);
//       cout << steps << endl;
//       continue;
//     }
//
//     if(cycle[v] == 0) {
//       cout << -1 <<endl;
//       continue;
//     }
//   }
// }
//
// int main() {
//   ios::sync_with_stdio(false);
//   cin.tie(nullptr);
//   int t = 1;
//   //cin >> t;
//   while(t--)
//     solve();
// }
```

Hungarian

```
#include <bits/stdc++.h>

using namespace std;

#define fst first
#define snd second
#define ll long long
#define ms(a, b) memset(a, b, sizeof(a))

const int INF = 1 << 30;
const int MAXN = 25;
const int MAXM = 25;

// hungarian weighted matching algo
// finds the MAX cost of MAX matching, to find mincost, add
edges as negatives
// Nodes are indexed from 1 on both sides
template<typename T>
struct KuhnMunkras { // n for left, m for right
  int n, m, match[MAXM];
  T g[MAXN][MAXM], lx[MAXN], ly[MAXM], slack[MAXM];
  bool vx[MAXN], vy[MAXM];

  void init(int n_, int m_) {
    ms(g,0); n = n_, m = m_;
  }

  void add(int u, int v, T w) {
```

```
    g[u][v] = w;
  }

  bool find(int x) {
    vx[x] = true;
    for (int y = 1; y <= m; ++y) {
      if (!vy[y]) {
        T delta = lx[x] + ly[y] - g[x][y];
        if (delta==0) {
          vy[y] = true;
          if (match[y] == 0 || find(match[y])) {
            match[y] = x;
            return true;
          }
        } else slack[y] = min(slack[y], delta);
      }
    }
    return false;
  }

  T matching() { // maximum weight matching
    fill(lx + 1, lx + 1 + n, numeric_limits<T>::lowest());
    ms(ly,0);
    ms(match,0);
    for (int i = 1; i <= n; ++i) {
      for (int j = 1; j <= m; ++j) lx[i] = max(lx[i], g[i]
[j]);
    }
    for (int k = 1; k <= n; ++k) {
      fill(slack + 1, slack + 1 + m,
numeric_limits<T>::max());
      while (true) {
        ms(vx,0);
        ms(vy,0);
        if (find(k)) break;
        else {
          T delta = numeric_limits<T>::max();
          for (int i = 1; i <= m; ++i) {
            if (!vy[i]) delta = min(delta, slack[i]);
          }
          for (int i = 1; i <= n; ++i) {
            if (vx[i]) lx[i] -= delta;
          }
          for (int i = 1; i <= m; ++i) {
            if (vy[i]) ly[i] += delta;
            if (!vy[i]) slack[i] -= delta;
          }
        }
      }
    }
    T result = 0;
    for (int i = 1; i <= n; ++i) result += lx[i];
    for (int i = 1; i <= m; ++i) result += ly[i];
    return result;
  }
};

// Ejemplo de uso anterior (comentado para pegar el algoritmo
con la plantilla):
// void solve() {
//   int n, m;
//   cin >> n >> m;
//
//   KuhnMunkras<double> km;
//   km.init(n, m);
//
//   for (int i = 1; i <= n; ++i) {
//     for (int j = 1; j <= m; ++j) {
//       double c;
//       cin >> c;
//       km.add(i,j, -c);
//     }
//   }
//
//   double result = km.matching();
//   double total_cost = -result;
//   cout << total_cost << '\n';
// }
//
// int main() {
//   ios::sync_with_stdio(false);
//   cin.tie(nullptr);
//   int t = 1;
//   //cin >> t;
//   while(t--)
//     solve();
// }
```

Kosaraju Scc

```
#include <bits/stdc++.h>

using namespace std;

#define fst first
#define snd second
#define ll long long
#define ms(a, b) memset(a, b, sizeof(a))

const int INF = 1 << 30;
const int MAXN = 25;
const int MAXM = 25;

vector<bool> visited; // keeps track of which vertices are
already visited

// runs depth first search starting at vertex v.
// each visited vertex is appended to the output vector when
dfs leaves it.
void dfs(int v, vector<vector<int>> const& adj, vector<int>
&output) {
  visited[v] = true;
  for (auto u : adj[v])
    if (!visited[u])
      dfs(u, adj, output);
  output.push_back(v);
}

// input: adj -- adjacency list of G
// output: components -- the strongly connected components in
G
// output: adj_cond -- adjacency list of G^SCC (by root
vertices)
void strongly_connected_components(vector<vector<int>> const&
adj,
    vector<vector<int>> &components,
    vector<vector<int>> &adj_cond) {
  int n = (int)adj.size();
  components.clear();
  adj_cond.clear();

  // 1) Order the vertices by finish time in a first DFS.
  vector<int> order;
  visited.assign(n, false);
  for (int i = 0; i < n; i++) {
    if (!visited[i]) {
      dfs(i, adj, order);
    }
  }

  // 2) Reverse all edges to get the transpose graph.
  vector<vector<int>> adj_rev(n);
  for (int v = 0; v < n; v++) {
    for (int u : adj[v]) {
      adj_rev[u].push_back(v);
    }
  }

  // 3) Do a second DFS in descending order of finish times
(from step 1).
  visited.assign(n, false);
  reverse(order.begin(), order.end());

  // this array will store the "root" (representative) of each
SCC
  vector<int> roots(n, -1);

  for (auto v : order) {
    if (!visited[v]) {
      vector<int> component;
      dfs(v, adj_rev, component);
      // store the newly found component
      components.push_back(component);
      // the root is the minimum vertex in that component
(arbitrary choice)
      int root = *min_element(begin(component),
end(component));
      for (auto u : component) {
        roots[u] = root;
      }
    }
  }

  // 4) Build the condensation graph (SCC DAG).
  adj_cond.assign(n, {});
  for (int v = 0; v < n; v++) {
```

```
    for (auto u : adj[v]) {
      if (roots[v] != roots[u]) {
        adj_cond[roots[v]].push_back(roots[u]);
      }
    }
  }
}

// Ejemplo de uso anterior (comentado para pegar el algoritmo
con la plantilla):
// void solve() {
//
//   int n = 5, m = 5;
//   vector<vector<int>> adj(n);
//
//   vector<pair<int,int>> edges = {
//     {0,1},
//     {1,2},
//     {2,0},
//     {1,3},
//     {3,4}
//   };
//
//   for (auto &e : edges) {
//     adj[e.first].push_back(e.second);
//   }
//
//   vector<vector<int>> components, cond_graph;
//   strongly_connected_components(adj, components,
cond_graph);
//
//   for (auto &comp : components) {
//     sort(comp.begin(), comp.end());
//   }
//
//   cout << "Number of SCCs: " << (int)components.size() <<
"\n";
//   for (int i = 0; i < (int)components.size(); i++) {
//     cout << "SCC " << i << ": ";
//     for (auto &v : components[i]) {
//       cout << v << " ";
//     }
//     cout << "\n";
//   }
//
//   cout << "Condensation graph adjacency list:\n";
//   // We'll print only non-empty adjacency to see edges
//   for (int i = 0; i < (int)cond_graph.size(); i++) {
//     if (!cond_graph[i].empty()) {
//       cout << "root " << i << ": ";
//       for (auto &v : cond_graph[i]) {
//         cout << v << " ";
//       }
//       cout << "\n";
//     }
//   }
// }
//
// int main() {
//   ios::sync_with_stdio(false);
//   cin.tie(nullptr);
//   int t = 1;
//   //cin >> t;
//   while(t--)
//     solve();
// }
```

Kruskal

```
#include <bits/stdc++.h>
using namespace std;

#define ll long long
#define endl '\n'

struct Edge {
  ll u, v, weight;
  bool operator<(Edge const& other) const {
    return weight < other.weight;
  }
};

struct DSU {
  vector<int> parent, size;

  DSU(int n) {
    parent.resize(n);
    size.assign(n, 1);
```

```
    iota(parent.begin(), parent.end(), 0);
  }

  int find(int v) {
    if (v == parent[v]) return v;
    return parent[v] = find(parent[v]);
  }

  bool unite(int a, int b) {
    a = find(a);
    b = find(b);
    if (a == b) return false;
    if (size[a] < size[b]) swap(a, b);
    parent[b] = a;
    size[a] += size[b];
    return true;
  }
};

ll kruskal(int n, vector<Edge>& edges, vector<Edge>& result) {
  sort(edges.begin(), edges.end());
  DSU dsu(n);
  ll cost = 0;

  for (auto& e : edges) {
    if (dsu.unite(e.u, e.v)) {
      cost += e.weight;
      result.push_back(e);
    }
  }
  return cost;
}

// Ejemplo de uso anterior (comentado para pegar el algoritmo
con la plantilla):
// int main() {
//   ios::sync_with_stdio(false);
//   cin.tie(nullptr);
//
//   int n, m;
//   cin >> n >> m;
//
//   vector<Edge> edges;
//   edges.reserve(m);
//
//   for (int i = 0; i < m; i++) {
//     ll u, v, w;
//     cin >> u >> v >> w;
//     u--, v--;
//     edges.push_back({u, v, w});
//   }
//
//   vector<Edge> result;
//   ll total_cost = kruskal(n, edges, result);
//
//   if(result.size() == n-1) {
//     cout << total_cost << endl;
//   } else {
//     cout << "IMPOSSIBLE" << endl;
//   }
//
//   // Optional: print MST edges
//   // for (auto &e : result)
//   //     cout << e.u + 1 << " " << e.v + 1 << " " <<
e.weight << endl;
// }
```

Topological Sort

```
#include <bits/stdc++.h>

using namespace std;

#define fst first
#define snd second
#define ll long long

const int INF = 1 << 30;
const int MAXN = 1e5+5;

vector<int> adj[MAXN];
vector<int> topo;

int n,m;

void dfs(int u, vector<int> &visited) {
  visited[u] = 1;
  for (int &v : adj[u]) {
```

```
    if (!visited[v])
      dfs(v, visited);
    else if(visited[v] == 1){
      cout << "IMPOSSIBLE\n";
      exit(0);
    }
  }
  visited[u] = 2;
  topo.push_back(u);
}

void topological_sort() {
  topo.clear();
  vector<int> visited(n+1,0);

  for (int i = 1; i <= n; i++)
    if (!visited[i])
      dfs(i, visited);

  reverse(topo.begin(), topo.end());
}

// Ejemplo de uso anterior (comentado para pegar el algoritmo
con la plantilla):
// void solve() {
//   cin >> n >> m;
//   for(int i = 0; i <= n; i++) adj[i].clear();
//   while(m--) {
//     int a,b;
//     cin >> a >> b;
//     adj[a].push_back(b);
//   }
//   topological_sort();
//   for (int x : topo) cout << x << " ";
//   cout << "\n";
// }
//
// int main() {
//   int t = 1;
//   // cin >> t;
//   while(t--)
//     solve();
// }
```

Math

Binary Exponentiation

```
#include <bits/stdc++.h>

#define ll long long

ll binpow(ll a, ll b) {
    if (b == 0)
        return 1;
    ll res = binpow(a, b / 2);
    if (b % 2)
        return res * res * a;
    else
        return res * res;
}

ll binpowIt(ll a, ll b) {
    ll res = 1;
    while (b > 0) {
        if (b & 1)
            res = res * a;
        a = a * a;
        b >>= 1;
    }
    return res;
}

int power(ll x, ll y, ll M)
{
    if (y == 0)
        return 1;

    int p = power(x, y / 2, M) % M;
    p = (p * p) % M;

    return (y % 2 == 0) ? p : (x * p) % M;
}
// inverse modular power(A, M - 2, M)
```

Expected Value

```
// Expected Value & Probability (Esperanza Matematica)
// 1. Linealidad de la Esperanza: E[X + Y] = E[X] + E[Y]
(siempre valido, aun dependientes)
// 2. Variable indicadora: E[I_A] = P(A)
// 3. Ensayos geometricos (primer exito con prob p): E = 1/p
// 4. Coupon Collector (recolectar n cupones): E = n *
sum_{i=1}^n (1/i) = n * H_n

// ─── Sistema Lineal para Esperanza en Grafos de Estados /
Markov ──────────────
// E[u] = 1 + sum_{v} P(u -> v) * E[v]
// Se transforma en: E[u] - sum_{v} P(u -> v) * E[v] = 1 (con
E[target] = 0)
// Resuelto con Eliminacion Gaussiana O(N^3)

const ld EPS = 1e-11;

// Retorna 1 si tiene solucion unica, 0 si no o infinitas.
Solucion en ans.
int gauss(vector<vector<ld>> a, vector<ld>& ans) {
    int n = sz(a), m = sz(a[0]) - 1;
    vi where(m, -1);
    for (int col = 0, row = 0; col < m && row < n; ++col) {
        int sel = row;
        for (int i = row; i < n; ++i)
            if (fabsl(a[i][col]) > fabsl(a[sel][col])) sel =
i;
        if (fabsl(a[sel][col]) < EPS) continue;
        for (int i = col; i <= m; ++i) swap(a[sel][i], a[row]
[i]);
        where[col] = row;

        for (int i = 0; i < n; ++i) {
            if (i != row) {
                ld c = a[i][col] / a[row][col];
                for (int j = col; j <= m; ++j) a[i][j] -=
a[row][j] * c;
            }
        }
        ++row;
    }
    ans.assign(m, 0);
    rep(i, m) {
        if (where[i] != -1) ans[i] = a[where[i]][m] /
a[where[i]][i];
    }
    rep(i, n) {
        ld sum = 0;
        rep(j, m) sum += ans[j] * a[i][j];
        if (fabsl(sum - a[i][m]) > EPS) return 0;
    }
    return 1;
}

// Construccion de matriz para n estados (0..n-1), absorbiendo
en target
// adj[u] = lista de {v, prob}
vector<ld> expectedSteps(int n, int target, const
vector<vector<pair<int, ld>>>& adj) {
    vector<vector<ld>> A(n, vector<ld>(n + 1, 0));
    rep(u, n) {
        if (u == target) {
            A[u][u] = 1;
            A[u][n] = 0; // E[target] = 0
            continue;
        }
        A[u][u] = 1.0;
        A[u][n] = 1.0; // sumatoria de costo/paso base = 1
        for (auto& edge : adj[u]) {
            int v = edge.ff;
            ld p = edge.ss;
            A[u][v] -= p;
        }
    }
    vector<ld> E;
    gauss(A, E);
    return E;
}
```

FFT

```
#include <bits/stdc++.h>
using namespace std;

#define ll long long
const double PI = acos(-1);
```

```
namespace FFT {
    static vector<int> rev;
    static vector<complex<double>> raices{{0,0},{1,0}};

    void fft(vector<complex<double>>& a, bool invertido) {
        int n = a.size();

        if (rev.size() != n) {
            int k = __builtin_ctz(n); // log2(n)
            rev.assign(n, 0);
            for (int i = 0; i < n; i++) {
                rev[i] = (rev[i >> 1] >> 1) | ((i & 1) << (k -
1));
            }
        }

        for (int i = 0; i < n; i++) {
            if (i < rev[i]) swap(a[i], a[rev[i]]);
        }

        if (raices.size() < n) {
            int k = __builtin_ctz(raices.size());
            raices.resize(n);
            while ((1 << k) < n) {
                double angulo = 2 * PI / (1 << (k + 1));
                for (int i = 1 << (k - 1); i < (1 << k); i++)
{
                    raices[2 * i] = raices[i];
                    double ang = angulo * (2 * i + 1 - (1 <<
k));
                    raices[2 * i + 1] = complex<double>
(cos(ang), sin(ang));
                }
                k++;
            }
        }

        for (int len = 1; len < n; len = len << 1) {
            for (int i = 0; i < n; i += 2 * len) {
                for (int j = 0; j < len; j++) {
                    complex<double> u = a[i + j];
                    complex<double> v = a[i + j + len] *
raices[len + j];
                    a[i + j] = u + v;
                    a[i + j + len] = u - v;
                }
            }
        }

        if (invertido) {
            reverse(a.begin() + 1, a.end());
            for (auto &x : a) x /= n;
        }
    }
} // namespace FFT

vector<ll> multiplicacion_polinomio(const vector<ll>& A, const
vector<ll>& B) {
    int n1 = A.size(), n2 = B.size();
    if (n1 == 0 || n2 == 0) return vector<ll>();

    int n = 1;
    while (n < n1 + n2 - 1) n = n << 1;
    vector<complex<double>> fa(n), fb(n);
    for (int i = 0; i < n1; i++) fa[i] = complex<double>
(A[i],0);
    for (int i = 0; i < n2; i++) fb[i] = complex<double>
(B[i],0);

    FFT::fft(fa, false);
    FFT::fft(fb, false);
    for (int i = 0; i < n; i++) fa[i] *= fb[i];
    FFT::fft(fa, true);

    vector<ll> res(n1 + n2 - 1);
    for (int i = 0; i < res.size(); i++) {
        res[i] = llround(fa[i].real()); // Redondeo
    }
    return res;
}

// Ejemplo de uso anterior (comentado para pegar el algoritmo
con la plantilla):
// int main() {
//     vector<ll> A = {1, 2, 3}; // 1 + 2x + 3x^2
//     vector<ll> B = {2, 0, 1}; // 2 + 0x + 1x^2
//     vector<ll> C = multiplicacion_polinomio(A, B);
```

```
//     return 0;
// }
```

Matrix Exponentiation

```
#include <bits/stdc++.h>

using namespace std;

#define ll long long
const ll MOD = 998244353;

struct Matrix
{
  int rows, cols;
  vector<vector<ll>> mat;

  // Matriz rectangular
  Matrix(int r, int c) : rows(r), cols(c), mat(r, vector<ll>
(c, 0)) {}

  // Matriz cuadrada
  Matrix(int n) : rows(n), cols(n), mat(n, vector<ll>(n, 0))
{}

  // Identity matrix (matriz cuadrada)
  static Matrix identity(int n)
  {
    Matrix I(n);
    for (int i = 0; i < n; i++)
      I.mat[i][i] = 1;
    return I;
  }

  // Matrix multiplication (permite matrices rectangulares)
  Matrix operator*(const Matrix &other) const
  {
    Matrix product(rows, other.cols);
    for (int i = 0; i < rows; ++i)
      for (int k = 0; k < cols; ++k)
      {
        ll val = mat[i][k] % MOD;
        for (int j = 0; j < other.cols; ++j)
        {
          product.mat[i][j] = (product.mat[i][j] + val *
other.mat[k][j]) % MOD;
        }
      }
    return product;
  }

  // Matrix exponentiation (solo matrices cuadradas rows ==
cols)
  Matrix pow(ll power) const
  {
    Matrix result = identity(rows);
    Matrix base = *this;

    while (power > 0)
    {
      if (power & 1)
        result = result * base;
      base = base * base;
      power >>= 1;
    }
    return result;
  }
};

ll fibonacci(int n)
{
  if (n == 0)
    return 0;
  Matrix base(2);
  base.mat = {{1, 1}, {1, 0}};
  base = base.pow(n - 1);
  return base.mat[0][0]; // F(n)
}

// Ejemplo de uso anterior (comentado para pegar el algoritmo
con la plantilla):
// int main() {
//   for(int i = 0; i < 25; i++)
//     cout << fibonacci(i) << " ";
// }

// Para crear Funciones con Recurrencias de la forma: F(n) =
a1*F(n-1) + a2*F(n-2) + ... + ad*F(n-d)
```

```
// La matriz inicial es: A = {{F(1), F(2), ... , F(d)}}
/* La matriz clave es:
   B =
   {
   {0 , 0 , 0, ... , ad},
   {1 , 0 , 0, ... , ad-1},
   {0 , 1 , 0, ... , ad-2},
   {. , . , ., ... , a2},
   {0 , 0 , 0,  1  , a1}
   }

   Se realiza:
   B = B.pow(n-d);
   A = A * B;

   En A queda: A = {{F(n-(d-1)), F(n-(d-2)), ..., F(n)}}
   Entonces F(n) = A[0][d-1];
*/
```

NTT

```
#include <bits/stdc++.h>
using namespace std;

#define ll long long

namespace NTT {
    // MOD debe ser un primo de la forma c * 2^k + 1
    // 998244353 = 119 * 2^23 + 1. Raíz primitiva G = 3.
    const ll MOD = 998244353;
    const ll G = 3; // Raíz primitiva para MOD

    ll fast_pow(ll a, ll b) {
        ll res = 1;
        a %= MOD;
        while (b > 0) {
            if (b & 1) res = (res * a) % MOD;
            a = (a * a) % MOD;
            b >>= 1;
        }
        return res;
    }

    ll inv(ll a) {
        return fast_pow(a, MOD - 2);
    }

    static vector<int> rev;
    static vector<ll> w;

    void precompute(int n) {
        if (rev.size() == n) return;
        int k = __builtin_ctz(n);
        rev.assign(n, 0);
        for (int i = 0; i < n; i++) {
            rev[i] = (rev[i >> 1] >> 1) | ((i & 1) << (k -
1));
        }

        w.resize(n);
        w[0] = 1;
        ll g_n = fast_pow(G, (MOD - 1) / n);
        for(int i = 1; i < n; i++) {
            w[i] = (w[i - 1] * g_n) % MOD;
        }
    }

    void fft(vector<ll>& a, bool invertido) {
        int n = a.size();
        precompute(n);

        for (int i = 0; i < n; i++) {
            if (i < rev[i]) swap(a[i], a[rev[i]]);
        }

        for (int len = 1; len < n; len <<= 1) {
            int step = n / (2 * len);
            for (int i = 0; i < n; i += 2 * len) {
                for (int j = 0; j < len; j++) {
                    ll u = a[i + j];
                    ll v = (a[i + j + len] * w[j * step]) %
MOD;

                    a[i + j] = (u + v) % MOD;
                    a[i + j + len] = (u - v + MOD) % MOD;
                }
            }
        }
```

```
        if (invertido) {
            reverse(a.begin() + 1, a.end());
            ll n_inv = inv(n);
            for (auto &x : a) {
                x = (x * n_inv) % MOD;
            }
        }
    }
} // namespace NTT

vector<ll> multiplicacion_polinomio_mod(const vector<ll>& A,
const vector<ll>& B) {
    int n1 = A.size(), n2 = B.size();
    if (n1 == 0 || n2 == 0) return vector<ll>();

    int n = 1;
    while (n < n1 + n2 - 1) n = n << 1;

    vector<ll> fa(n), fb(n);
    for (int i = 0; i < n1; i++) fa[i] = (A[i] % NTT::MOD +
NTT::MOD) % NTT::MOD;
    for (int i = 0; i < n2; i++) fb[i] = (B[i] % NTT::MOD +
NTT::MOD) % NTT::MOD;

    NTT::fft(fa, false);
    NTT::fft(fb, false);

    for (int i = 0; i < n; i++) {
        fa[i] = (fa[i] * fb[i]) % NTT::MOD;
    }

    NTT::fft(fa, true);

    fa.resize(n1 + n2 - 1);
    return fa;
}

// Ejemplo de uso anterior (comentado para pegar el algoritmo
con la plantilla):
// int main() {
//     vector<ll> A = {1, 2, 3}; // 1 + 2x + 3x^2
//     vector<ll> B = {2, 0, 1}; // 2 + 0x + 1x^2
//     vector<ll> C = multiplicacion_polinomio_mod(A, B);
//     return 0;
// }
```

Pollard Rho

```
#include <bits/stdc++.h>
using namespace std;

using u64 = uint64_t;
using u128 = __uint128_t;

// Multiplicacion modular para int128
u64 binpower(u64 base, u64 e, u64 mod)
{
    u64 result = 1;
    base %= mod;
    while (e)
    {
        if (e & 1)
            result = (u128)result * base % mod;
        base = (u128)base * base % mod;
        e >>= 1;
    }
    return result;
}

// Test de primalidad de Miller Rabin
bool check(u64 n, u64 a, u64 d, int s)
{
    u64 x = binpower(a, d, n);
    if (x == 1 || x == n - 1)
        return false;
    for (int r = 1; r < s; r++)
    {
        x = (u128)x * x % n;
        if (x == n - 1)
            return false;
    }
    return true;
}
// Retorna true si es primo
bool MillerRabin(u64 n)
{
    if (n < 2)
```

```
        return false;
    int r = 0;
    u64 d = n - 1;
    while ((d & 1) == 0)
    {
        d >>= 1;
        r++;
    }
    // Bases suficientes para verificar primalidad hasta 2^64
    for (int a : {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37})
    {
        if (n == a)
            return true;
        if (check(n, a, d, r))
            return false;
    }
    return true;
}

// Algoritmo Pollard Rho, devuelve un factor primo de n
u64 pollard_rho(u64 n)
{
    if (n % 2 == 0)
        return 2;
    if (MillerRabin(n))
        return n;

    u64 x = 2, y = 2, d = 1, c = 1;

    // Función pseudoaleatoria f(x) = x^2 + c
    auto f = [&](u64 x, u64 n, u64 c)
    {
        return ((u128)x * x + c) % n;
    };

    while (d == 1)
    {
        x = f(x, n, c);          // Tortuga (1 paso)
        y = f(f(y, n, c), n, c); // Liebre (2 pasos)

        u64 diff = x > y ? x - y : y - x;
        d = std::gcd(diff, n);

        if (d == n)
        {
            x = rand() % (n - 2) + 2;
            y = x;
            c = rand() % (n - 1) + 1;
            d = 1;
        }
    }
    return d;
}

// Función recursiva para obtener todos los factores primos y
sus exponentes
void factorize(u64 n, map<u64, int> &factors)
{
    if (n == 1)
        return;
    if (MillerRabin(n))
    {
        factors[n]++;
        return;
    }
    u64 divisor = pollard_rho(n);
    factorize(divisor, factors);
    factorize(n / divisor, factors);
}

int main()
{
    ios_base::sync_with_stdio(false);
    cin.tie(NULL);

    u64 n;
    cin >> n;
    // factors, obtiene todos los factores y exponentes
    map<u64, int> factors;
    factorize(n, factors);
    return 0;
}
```

Sieve

```
// Sieve of Eratosthenes - O(N log log N)

// ─── 1. Criba Estandar (Arreglo Booleano)
```

```
_____________________________________
const int MAXN = 1e6 + 5;
bool is_prime[MAXN];
vi primes;

void sieve() {
    fill(is_prime, is_prime + MAXN, true);
    is_prime[0] = is_prime[1] = false;

    // Tachar pares primero
    for (int i = 4; i < MAXN; i += 2) is_prime[i] = false;

    // Solo revisar impares
    for (int i = 3; (ll)i * i < MAXN; i += 2) {
        if (!is_prime[i]) continue;
        for (int j = i * i; j < MAXN; j += 2 * i) {
            is_prime[j] = false;
        }
    }

    primes.pb(2);
    for (int i = 3; i < MAXN; i += 2) {
        if (is_prime[i]) primes.pb(i);
    }
}

// ─── 2. Criba con Bitset (Ultra Rapida, 8x menos memoria, N
hasta 1e7) ──────
const int MAX_BITSET = 10000005;
bitset<MAX_BITSET> bs;
vi primes_bs;

void bitsetSieve() {
    bs.set(); // Pone todos los bits en 1
    bs[0] = bs[1] = 0;
    for (int i = 4; i < MAX_BITSET; i += 2) bs[i] = 0;

    for (int i = 3; (ll)i * i < MAX_BITSET; i += 2) {
        if (bs[i]) {
            for (int j = i * i; j < MAX_BITSET; j += 2 * i) {
                bs[j] = 0;
            }
        }
    }

    primes_bs.pb(2);
    for (int i = 3; i < MAX_BITSET; i += 2) {
        if (bs[i]) primes_bs.pb(i);
    }
}
```

Strings

Aho Corasick

```
#include <bits/stdc++.h>
using namespace std;

// ALPHA is a compile-time constant: one contiguous allocation
for all nodes.
// toIdx maps a character to [0, ALPHA) — swap freely for any
alphabet.
template<int ALPHA>
struct AhoCorasick {
    function<int(char)> toIdx;
    vector<array<int, ALPHA>> go;
    vector<int> fail, dict, output;
    // output[v] = pattern index ending at v (-1 if none)

    AhoCorasick(function<int(char)> toIdx) : toIdx(toIdx) {
        go.emplace_back(); go.back().fill(-1);
        fail.push_back(0);
        dict.push_back(-1);
        output.push_back(-1);
    }

    int insert(const string& s, int patId) {
        int cur = 0;
        for (char c : s) {
            int ch = toIdx(c);
            if (go[cur][ch] == -1) {
                go[cur][ch] = go.size();
                go.emplace_back(); go.back().fill(-1);
                fail.push_back(0);
                dict.push_back(-1);
                output.push_back(-1);
            }
```

```
            cur = go[cur][ch];
        }
        output[cur] = patId;
        return cur;
    }

    // Build failure and dictionary links (call after all
insertions)
    void build() {
        queue<int> q;
        for (int c = 0; c < ALPHA; c++) {
            if (go[0][c] == -1) {
                go[0][c] = 0; // missing char at root loops
back to root
            } else {
                fail[go[0][c]] = 0;
                q.push(go[0][c]);
            }
        }
        while (!q.empty()) {
            int u = q.front(); q.pop();
            dict[u] = (output[fail[u]] != -1) ? fail[u] :
dict[fail[u]];
            for (int c = 0; c < ALPHA; c++) {
                if (go[u][c] == -1) {
                    go[u][c] = go[fail[u]][c]; // compressed
goto
                } else {
                    fail[go[u][c]] = go[fail[u]][c];
                    q.push(go[u][c]);
                }
            }
        }
    }

    void search(const string& text, const vector<string>&
pats,
                function<void(int,int)> cb) {
        int cur = 0;
        for (int i = 0; i < (int)text.size(); i++) {
            cur = go[cur][toIdx(text[i])];
            for (int v = cur; v > 0; v = dict[v])
                if (output[v] >= 0)
                    cb(i - (int)pats[output[v]].size() + 1,
output[v]);
        }
    }

};

// Ejemplo de uso anterior (main comentado para compilar con
la plantilla):
// int main() {
//   vector<string> patterns = {"he", "she", "his", "hers"};
//   AhoCorasick<26> ac([](char c) { return c - 'a'; });
//   for (int i = 0; i < (int)patterns.size(); ++i)
ac.insert(patterns[i], i);
//   ac.build();
//   ac.search("ahishers", patterns, [&](int pos, int id) {
//     cout << patterns[id] << " at " << pos << "\n";
//   });
// }
```

Bit Trie

```
struct BitTrie {
  static const int B = 30; // bits (use 62 with ll values)
  struct Node {
    int nxt[2] = {-1, -1}, cnt = 0;
  };
  vector<Node> t;
  BitTrie() : t(1) {}
  void insert(int v, int d = 1) { // d = 1 insert, d = -1
erase (the value must exist)
    int x = 0;
    for (int b = B-1; b >= 0; b--) {
      int k = v>>b & 1;
      if (t[x].nxt[k] < 0) { t[x].nxt[k] = t.size();
t.emplace_back(); }
      x = t[x].nxt[k];
      t[x].cnt += d;
    }
  }
  int maxXor(int v) { // max of v ^ u over u in the set (set
must be non-empty)
    int x = 0, res = 0;
    for (int b = B-1; b >= 0; b--) {
      int k = v>>b & 1;
```

```
      int y = t[x].nxt[k^1];
      if (y >= 0 && t[y].cnt > 0) { res |= 1<<b; x = y; }
      else x = t[x].nxt[k];
    }
    return res;
  }
  int countLess(int v, int lim) { // number of u in the set
with (v ^ u) < lim
    int x = 0, res = 0;
    for (int b = B-1; b >= 0 && x >= 0; b--) {
      int k = v>>b & 1, l = lim>>b & 1;
      if (l) {
        int y = t[x].nxt[k];
        if (y >= 0) res += t[y].cnt; // xor bit 0 < 1, whole
subtree counts
        x = t[x].nxt[k^1];
      } else x = t[x].nxt[k];
    }
    return res;
  }
};
```

Double Hash

```
#include <bits/stdc++.h>
using namespace std;

// Double hashing: use two independent hash functions
// to make collision probability negligible (~1/P1 * 1/P2)

struct DoubleHash {
    static const long long MOD1 = 1e9 + 7;
    static const long long MOD2 = 1e9 + 9;
    static const long long B1   = 131;
    static const long long B2   = 137;

    vector<long long> h1, h2, pw1, pw2;
    int n;

    DoubleHash(const string& s) : n(s.size()),
        h1(s.size()+1,0), h2(s.size()+1,0),
        pw1(s.size()+1,1), pw2(s.size()+1,1)
    {
        for (int i = 0; i < n; i++) {
            h1[i+1] = (h1[i] * B1 + s[i]) % MOD1;
            h2[i+1] = (h2[i] * B2 + s[i]) % MOD2;
            pw1[i+1] = pw1[i] * B1 % MOD1;
            pw2[i+1] = pw2[i] * B2 % MOD2;
        }
    }

    // Returns a pair (hash1, hash2) for s[l..r]
    pair<long long, long long> query(int l, int r) const {
        long long q1 = (h1[r+1] - h1[l] * pw1[r-l+1] % MOD1 +
MOD1) % MOD1;
        long long q2 = (h2[r+1] - h2[l] * pw2[r-l+1] % MOD2 +
MOD2) % MOD2;
        return {q1, q2};
    }

    bool equal(int l1, int r1, int l2, int r2) const {
        return query(l1, r1) == query(l2, r2);
    }
};

// Ejemplo de uso anterior (main comentado para compilar con
la plantilla):
// int main() {
//   string s = "abracadabra";
//   DoubleHash h(s);
//   cout << h.query(0, 2).first << " " << h.query(0,
2).second << "\n";
//   cout << boolalpha << h.equal(0, 2, 7, 9) << "\n";
// }
```

Hash

```
#include <bits/stdc++.h>
using namespace std;

// Single polynomial hash of a string
// hash(s) = s[0]*B^(n-1) + s[1]*B^(n-2) + ... + s[n-1]*B^0
(mod P)
// Equivalent iterative: hash = hash*B + s[i]

const long long MOD = 1e9 + 7;
const long long BASE = 131;
```

```
long long hash_str(const string& s) {
    long long h = 0;
    for (char c : s) {
        h = (h * BASE + c) % MOD;
    }
    return h;
}

// Precompute prefix hashes and powers for O(1) substring
queries
struct HashTable {
    vector<long long> h, pw;
    long long mod, base;

    HashTable(const string& s, long long B = BASE, long long M
= MOD)
        : h(s.size() + 1, 0), pw(s.size() + 1, 1), mod(M),
base(B)
    {
        for (int i = 0; i < (int)s.size(); i++) {
            h[i + 1] = (h[i] * base + s[i]) % mod;
            pw[i + 1] = pw[i] * base % mod;
        }
    }

    // Hash of s[l..r] (0-indexed, inclusive)
    long long query(int l, int r) const {
        return (h[r + 1] - h[l] * pw[r - l + 1] % mod + mod) %
mod;
    }
};

// Ejemplo de uso anterior (main comentado para compilar con
la plantilla):
// int main() {
//   string s = "competitive";
//   HashTable h(s);
//   cout << hash_str(s) << "\n";
//   cout << h.query(0, 3) << "\n"; // hash of s[0..3]
// }
```

KMP Automaton

```
#include <bits/stdc++.h>

using namespace std;

#define ll long long
#define endl '\n'

vector<int> prefix_function(string s) {
  int n = (int)s.length();
  vector<int> pi(n);
  for (int i = 1; i < n; i++) {
    int j = pi[i-1];
    while (j > 0 && s[i] != s[j])
      j = pi[j-1];
    if (s[i] == s[j])
      j++;
    pi[i] = j;
  }
  return pi;
}

void compute_automaton(string s, vector<vector<int>>& aut) {
  s += '#';
  int n = s.size();
  vector<int> pi = prefix_function(s);
  aut.assign(n, vector<int>(26));
  for (int i = 0; i < n; i++) {
    for (int c = 0; c < 26; c++) {
      if (i > 0 && 'a' + c != s[i])
        aut[i][c] = aut[pi[i-1]][c];
      else
        aut[i][c] = i + ('a' + c == s[i]);
    }
  }
}

const int MAXN = 1e4+5;
const int MAXK = 1e3+5;
const int MOD = 1e9+7;


int n,k;
```

```
string pattern;
vector<vector<int>> aut;

long long binpow(long long a, long long b)
{
    long long result = 1;
    while (b) {
        if (b & 1)
            result = (result * a) % MOD;
        a = (a * a) % MOD;
        b >>= 1;
    }
    return result;
}

ll dp[MAXN][MAXK];
ll fdp(int pos, int state) {
  bool found = state == pattern.size();
  if(pos == n) {
    return  found;
  }

  ll &val = dp[pos][state];
  if(val != -1) return val;

  val = 0;

  if(found) {
    ll l = n-pos;
    return val = binpow(k, l);
  }

  for(int i = 0; i < k; i++) {
    val = (val + fdp(pos+1,aut[state][i])) % MOD;
  }

  return val;
}

// Ejemplo de uso anterior (comentado para pegar el algoritmo
con la plantilla):
// void solve() {
//   memset(dp, -1, sizeof dp);
//   cin >> n >> k >> pattern;
//
//   compute_automaton(pattern, aut);
//
//   cout << fdp(0,0) << endl;
// }
//
// int main() {
//   solve();
// }
```

KMP

```
#include <bits/stdc++.h>
using namespace std;

vector<int> buildFail(const string& P) {
    int m = P.size();
    vector<int> fail(m, 0);
    for (int i = 1, j = 0; i < m; i++) {
        while (j > 0 && P[i] != P[j]) j = fail[j - 1];
        if (P[i] == P[j]) j++;
        fail[i] = j;
    }
    return fail;
}

// ── Application 1: Count non-overlapping occurrences
──────────────────────
int countNonOverlapping(const string& T, const string& P) {
    auto fail = buildFail(P);
    int count = 0, j = 0, m = P.size();
    for (char c : T) {
        while (j > 0 && c != P[j]) j = fail[j - 1];
        if (c == P[j]) j++;
        if (j == m) { count++; j = 0; } // reset instead of
fail[m-1]
    }
    return count;
}

// ── Application 2: Shortest period of a string
──────────────────────────
// The period of S is the smallest p such that S[i] == S[i %
p] for all i.
```

```
// Using KMP: period = m - fail[m-1]   (if m % (m-fail[m-1])
== 0, else period = m)
int period(const string& S) {
    auto fail = buildFail(S);
    int m = S.size();
    int p = m - fail[m - 1];
    return (m % p == 0) ? p : m;
}

// —— Application 3: Concatenation search (good for multiple
patterns) —————————
// Search for P in T by building the "combined" string P + '#'
+ T,
// then checking where fail[i] == m in the combined string.
vector<int> kmpConcat(const string& T, const string& P) {
    string S = P + '#' + T;     // '#' is a separator not in
the alphabet
    auto fail = buildFail(S);
    int m = P.size();
    vector<int> matches;
    for (int i = m + 1; i < (int)S.size(); i++)
        if (fail[i] == m)
            matches.push_back(i - 2 * m); // convert back to
T-index
    return matches;
}

// Ejemplo de uso anterior (main comentado para compilar con
la plantilla):
// int main() {
//   string text = "ababa", pattern = "aba";
//   cout << countNonOverlapping(text, pattern) << "\n";
//   cout << period(text) << "\n";
//   for (int pos : kmpConcat(text, pattern)) cout << pos << "
";
//   cout << "\n";
// }
```

Rolling Hash

```
#include <bits/stdc++.h>
using namespace std;

const long long MOD = 1e9 + 7;
const long long BASE = 131;

long long power(long long base, long long exp, long long mod)
{
    long long result = 1;
    base %= mod;
    while (exp > 0) {
        if (exp & 1) result = result * base % mod;
        base = base * base % mod;
        exp >>= 1;
    }
    return result;
}

// Compute hash of every substring of length k in O(n) total
vector<long long> rolling_hash(const string& s, int k) {
    int n = s.size();
    if (n < k) return {};

    vector<long long> result;
    long long h = 0;
    long long base_k = power(BASE, k - 1, MOD);  // B^(k-1)

    // Initial window [0, k-1]
    for (int i = 0; i < k; i++)
        h = (h * BASE + s[i]) % MOD;
    result.push_back(h);

    for (int i = 1; i + k - 1 < n; i++) {
        // Remove s[i-1], add s[i+k-1]
        h = (h - (long long)s[i - 1] * base_k % MOD + MOD) %
MOD;
        h = (h * BASE + s[i + k - 1]) % MOD;
        result.push_back(h);
    }

    return result;
}

// Find all occurrences of pattern p in text t using rolling
hash
vector<int> rabin_karp(const string& t, const string& p) {
    int n = t.size(), m = p.size();
    long long ph = 0;
```

```
    for (char c : p) ph = (ph * BASE + c) % MOD;

    vector<long long> th = rolling_hash(t, m);
    vector<int> matches;
    for (int i = 0; i < (int)th.size(); i++)
        if (th[i] == ph)
            if (t.substr(i, m) == p)  // confirm match (avoid
hash collision)
                matches.push_back(i);
    return matches;
}

// Ejemplo de uso anterior (main comentado para compilar con
la plantilla):
// int main() {
//   string text = "abacaba", pattern = "aba";
//   for (int pos : rabin_karp(text, pattern)) cout << pos <<
" ";
//   cout << "\n";
// }
```

String Hashing

```
#include <bits/stdc++.h>

using namespace std;

#define ll long long

const int MAXN = 1e5+5;
const int BASE = 31;
const int M = 1e9 + 7;

int pot[MAXN];

void compute_powers(int n) {
    int p_pow = 1;
    for (int i = 0; i < n; i++) {
        pot[i] = p_pow;
        p_pow = (p_pow * BASE) % M;
    }
}

vector<ll> compute_hash(string const& s) {
    int n = s.size();
    vector<ll> hash_values(n + 5);
    hash_values[0] = 0;
    for (int i = 1; i <= n; i++) {
        hash_values[i] = (hash_values[i - 1] * BASE + s[i -
1]) % M;
    }
    return hash_values;
}

ll get_hash(vector<ll> &hash_values, int a, int b) {
    return (hash_values[b] - (hash_values[a - 1] * pot[b - a +
1]) % M + M) % M;
}

// Ejemplo de uso anterior (comentado para pegar el algoritmo
con la plantilla):
// int main() {
//     string word = "ALLEY";
//     compute_powers(word.size());
//     vector<ll> hash = compute_hash(word);
//
//     for (int i = 1; i <= word.size(); i++) {
//         cout << hash[i] << ' ';
//     }
//     cout << endl;
//     for (int i = 1; i <= word.size(); i++) {
//         cout << word[i - 1] << ": " << get_hash(hash, i, i)
<< endl;
//     }
//
//     return 0;
// }
```

Suffix Array

```
#include <bits/stdc++.h>
using namespace std;

vector<int> suffix_array(const string& s) {
    int n = s.size();
    vector<int> sa(n), rank_(n), tmp(n);
    iota(sa.begin(), sa.end(), 0);
    for (int i = 0; i < n; i++) rank_[i] = s[i];
```

```
    for (int k = 1; k < n; k <<= 1) {
        auto key = [&](int i) -> pair<int,int> {
            return {rank_[i], i + k < n ? rank_[i + k] : -1};
        };
        sort(sa.begin(), sa.end(), [&](int a, int b){ return
key(a) < key(b); });
        tmp[sa[0]] = 0;
        for (int i = 1; i < n; i++)
            tmp[sa[i]] = tmp[sa[i-1]] + (key(sa[i]) !=
key(sa[i-1]));
        rank_ = tmp;
        if (rank_[sa[n-1]] == n-1) break;
    }
    return sa;
}

vector<int> kasai_lcp(const string& s, const vector<int>& sa)
{
    int n = sa.size();
    vector<int> rank_(n), lcp(n, 0);
    for (int i = 0; i < n; i++) rank_[sa[i]] = i;
    int h = 0;
    for (int i = 0; i < n; i++) {
        if (rank_[i] > 0) {
            int j = sa[rank_[i] - 1];
            while (i + h < n && j + h < n && s[i+h] == s[j+h])
h++;
            lcp[rank_[i]] = h;
            if (h > 0) h--;
        }
    }
    return lcp;
}

// ── 1. Pattern search: O(M log N)
──────────────────────────────────────
// Binary search on SA: all matches are a contiguous range
[lo, hi).
pair<int,int> search_range(const string& s, const vector<int>&
sa, const string& P) {
    int n = sa.size(), m = P.size();
    int lo = (int)(lower_bound(sa.begin(), sa.end(), 0, [&]
(int a, int) {
        return s.compare(a, m, P) < 0;
    }) - sa.begin());
    int hi = (int)(upper_bound(sa.begin(), sa.end(), 0, [&]
(int, int a) {
        return s.compare(a, m, P) > 0;
    }) - sa.begin());
    return {lo, hi};
}

// ── 2. Count distinct substrings
──────────────────────────────────────
// Total substrings = N*(N+1)/2. Subtract duplicates counted
by LCP array.
long long distinct_substrings(const string& s) {
    auto sa  = suffix_array(s);
    auto lcp = kasai_lcp(s, sa);
    int n = s.size();
    long long total = (long long)n * (n + 1) / 2;
    for (int v : lcp) total -= v;
    return total;
}

// ── 3. Longest repeated substring
──────────────────────────────────────
// The maximum value in the LCP array.
string longest_repeated(const string& s) {
    auto sa  = suffix_array(s);
    auto lcp = kasai_lcp(s, sa);
    int best = *max_element(lcp.begin(), lcp.end());
    int idx  = max_element(lcp.begin(), lcp.end()) -
lcp.begin();
    return best > 0 ? s.substr(sa[idx], best) : "";
}

// Ejemplo de uso anterior (main comentado para compilar con
la plantilla):
// int main() {
//   string s = "banana";
//   vector<int> sa = suffix_array(s);
//   for (int pos : sa) cout << pos << " ";
//   cout << "\ndistinct substrings = " <<
distinct_substrings(s) << "\n";
//   cout << "longest repeated = " << longest_repeated(s) <<
```

```
"\n";
// }
```

Suffix Tree

```
#include <bits/stdc++.h>
using namespace std;
struct SuffixTree {
    struct Node {
        int ch[26], link, start, end, cnt;
        // end = INF for leaves (extends with each new char)
        Node(int s, int e) : link(0), start(s), end(e), cnt(0)
{
            fill(ch, ch + 26, -1);
        }
        int len(int phase) { return min(end, phase + 1) -
start; }
    };

    string s;
    vector<Node> t;
    int last, n;
    int activeNode, activeEdge, activeLen, remaining;
    int need;

    SuffixTree() {}

    void setSuffixLink(int v) {
        if (need != -1) t[need].link = v;
        need = v;
    }

    bool walkDown(int v) {
        if (activeLen >= t[v].len(n - 1)) {
            activeEdge += t[v].len(n - 1);
            activeLen  -= t[v].len(n - 1);
            activeNode  = v;
            return true;
        }
        return false;
    }

    void extend(int pos) {
        t[0].end = pos + 1;  // implicit: all leaves extend
        remaining++;
        need = -1;

        while (remaining > 0) {
            if (activeLen == 0) activeEdge = pos;
            int ae = s[activeEdge] - 'a';

            if (t[activeNode].ch[ae] == -1) {
                // Rule 2: create new leaf
                t[activeNode].ch[ae] = t.size();
                t.emplace_back(pos, INT_MAX);
                setSuffixLink(activeNode);
            } else {
                int nxt = t[activeNode].ch[ae];
                if (walkDown(nxt)) continue;

                if (s[t[nxt].start + activeLen] == s[pos]) {
                    // Rule 3: already exists — just extend
active len
                    activeLen++;
                    setSuffixLink(activeNode);
                    break;
                }

                // Rule 2: split
                int split = t.size();
                t.emplace_back(t[nxt].start, t[nxt].start +
activeLen);
                t[activeNode].ch[ae] = split;

                t[split].ch[s[pos] - 'a'] = t.size();
                t.emplace_back(pos, INT_MAX);
                t[nxt].start += activeLen;
                t[split].ch[s[t[nxt].start] - 'a'] = nxt;
                setSuffixLink(split);
            }

            remaining--;
            if (activeNode == 0 && activeLen > 0) {
                activeLen--;
                activeEdge = pos - remaining + 1;
            } else {
                activeNode = t[activeNode].link ?
t[activeNode].link : 0;
```

```
            }
        }
        n = pos + 1;
    }

    void build(const string& str) {
        s = str + "$";
        n = 0;
        t.clear();
        t.emplace_back(0, 0);  // root
        last = activeNode = activeLen = activeEdge = remaining
= 0;
        need = -1;
        for (int i = 0; i < (int)s.size(); i++) extend(i);
    }

    // DFS to collect suffix indices at leaves
    void dfs(int v, int depth, vector<int>& res) {
        bool leaf = true;
        for (int c = 0; c < 26; c++) {
            if (t[v].ch[c] == -1) continue;
            leaf = false;
            dfs(t[v].ch[c], depth + t[t[v].ch[c]].len(n - 1),
res);
        }
        if (leaf) {
            // leaf: suffix index = n - 1 - depth  (n includes
'$')
            res.push_back(n - 1 - depth);
        }
    }

    // Find all occurrences of pattern P in O(|P| + occ)
    vector<int> search(const string& p) {
        int v = 0, i = 0;
        while (i < (int)p.size()) {
            int c = p[i] - 'a';
            if (t[v].ch[c] == -1) return {};
            int nxt = t[v].ch[c];
            for (int j = t[nxt].start; j < t[nxt].end && i <
(int)p.size(); j++, i++)
                if (s[j] != p[i]) return {};
            v = nxt;
        }
        vector<int> res;
        dfs(v, 0, res);
        sort(res.begin(), res.end());
        return res;
    }
};

// --- Longest Repeated Substring ---
// DFS: track string depth (total chars from root).
// Deepest internal node (>= 2 children) gives LRS.
string longestRepeated(SuffixTree& st) {
    string best = "";
    function<void(int, int)> dfs = [&](int v, int depth) {
        int childCount = 0;
        for (int c = 0; c < 26; c++) {
            if (st.t[v].ch[c] == -1) continue;
            childCount++;
            int nxt = st.t[v].ch[c];
            dfs(nxt, depth + st.t[nxt].len(st.n - 1));
        }
        // internal node with depth > best
        if (childCount > 0 && depth > (int)best.size()) {
            // walk back up to find the string — simplified
via start pointer
            int start = st.t[v].start - (depth -
st.t[v].len(st.n - 1));
            best = st.s.substr(max(0, start), depth);
        }
    };
    dfs(0, 0);
    return best;  // removes trailing '$' if present
}

// --- Count distinct substrings ---
// Each edge contributes (edgeLen) new distinct substrings.
// Total = sum of all edge lengths in the tree.
long long countDistinct(SuffixTree& st) {
    long long total = 0;
    // node 0 is root (no parent edge), skip
    for (int v = 1; v < (int)st.t.size(); v++)
        total += st.t[v].len(st.n - 1);
    return total - 1;  // subtract 1 for the empty string
}
```

```
// --- Longest Common Substring of two strings ---
// Build suffix tree of S1 + '#' + S2 + '$'
// Find deepest internal node whose subtree has leaves from
both S1 and S2.
string longestCommon(const string& s1, const string& s2) {
    SuffixTree st;
    string combined = s1 + "#" + s2 + "$";
    st.build(combined);
    int n1 = s1.size();
    string best = "";

    // DFS: returns {has_s1_leaf, has_s2_leaf}
    function<pair<bool,bool>(int, int)> dfs = [&](int v, int
dep) -> pair<bool,bool> {
        bool has1 = false, has2 = false;
        bool leaf = true;

        for (int c = 0; c < 26; c++) {
            if (st.t[v].ch[c] == -1) continue;
            leaf = false;
            int nxt = st.t[v].ch[c];
            auto [h1, h2] = dfs(nxt, dep + st.t[nxt].len(st.n
- 1));
            has1 |= h1;
            has2 |= h2;
        }

        if (leaf) {
            int sufIdx = st.n - 1 - dep;
            has1 = (sufIdx < (int)n1);
            has2 = (sufIdx > (int)n1);  // > n1 skips the '#'
        }

        // internal node with leaves from both strings
        if (!leaf && has1 && has2 && dep > (int)best.size()) {
            // recover substring: edge ends at t[v].end, go
back 'dep' chars
            int endPos = st.t[v].end;
            int startPos = endPos - dep;
            if (startPos >= 0 && endPos <= (int)st.s.size())
                best = st.s.substr(startPos, dep);
        }
        return {has1, has2};
    };
    dfs(0, 0);
    return best;
}

// --- Pattern matching demo ---

// Ejemplo de uso anterior (main comentado para compilar con
la plantilla):
// int main() {
//   SuffixTree st;
//   st.build("banana");
//   cout << longestRepeated(st) << "\n";
//   cout << countDistinct(st) << "\n";
//   cout << longestCommon("banana", "ananas") << "\n";
//   for (int pos : st.search("ana")) cout << pos << " ";
//   cout << "\n";
// }
```

Trie

```
struct Trie {

  static const int A = 26; // alphabet size
  struct Node {
    int nxt[A], cnt = 0, end = 0; // cnt: words passing here,
end: words ending here
    Node() { memset(nxt, -1, sizeof nxt); }
  };
  vector<Node> t;
  Trie() : t(1) {}
  void insert(const string& s) {
    int x = 0;
    for (char c : s) {
      int k = c - 'a';
      if (t[x].nxt[k] < 0) { t[x].nxt[k] = t.size();
t.emplace_back(); }
      x = t[x].nxt[k];
      t[x].cnt++;
    }
    t[x].end++;
  }
  int find(const string& s) { // node of s, or -1
    int x = 0;
```

```
    for (char c : s) {
      x = t[x].nxt[c - 'a'];
      if (x < 0) return -1;
    }
    return x;
  }
  int count(const string& s) { int x = find(s); return x < 0 ?
0 : t[x].end; }  // exact matches
  int prefix(const string& s) { int x = find(s); return x < 0
? 0 : t[x].cnt; } // words with prefix s
  bool erase(const string& s) { // removes one copy, false if
absent
    if (!count(s)) return false;
    int x = 0;
    for (char c : s) { x = t[x].nxt[c - 'a']; t[x].cnt--; }
    t[x].end--;
    return true;
  }
};
```

Z Function

```
#include <bits/stdc++.h>
using namespace std;

vector<int> z_function(const string& s) {
    int n = s.size();
    vector<int> z(n, 0);
    z[0] = n;
    int l = 0, r = 0;
    for (int i = 1; i < n; i++) {
        if (i < r) z[i] = min(z[i - l], r - i);
        while (i + z[i] < n && s[z[i]] == s[i + z[i]]) z[i]++;
        if (i + z[i] > r) { l = i; r = i + z[i]; }
    }
    return z;
}

// ── 1. Pattern matching: find all occurrences of P in T
──────────────────
// Concatenate P + '#' + T. Any position i in the T-part where
z[i] == |P|
// corresponds to a match at T[i - |P| - 1].
vector<int> find_all(const string& P, const string& T) {
    string s = P + '#' + T;
    auto z = z_function(s);
    int m = P.size();
    vector<int> matches;
    for (int i = m + 1; i < (int)s.size(); i++)
        if (z[i] == m) matches.push_back(i - m - 1);
    return matches;
}

// ── 2. Shortest period of a string
──────────────────────────────────────
// The shortest period p satisfies: s[0..n-p-1] == s[p..n-1],
// i.e. z[p] + p == n (the suffix at p matches the prefix and
reaches the end).
int shortest_period(const string& s) {
    int n = s.size();
    auto z = z_function(s);
    for (int p = 1; p < n; p++)
        if (z[p] + p == n) return p;
    return n; // no period shorter than n
}

// ── 3. Longest border (prefix = suffix, proper)
──────────────────────────
// A border of length k exists iff z[n-k] == k.
int longest_border(const string& s) {
    int n = s.size();
    auto z = z_function(s);
    for (int k = n - 1; k >= 1; k--)
        if (z[n - k] == k) return k;
    return 0;
}

// Ejemplo de uso anterior (main comentado para compilar con
la plantilla):
// int main() {
//   string text = "abacaba", pattern = "aba";
//   for (int pos : find_all(pattern, text)) cout << pos << "
";
//   cout << "\nperiod = " << shortest_period(text) << "\n";
// }
```

Otros

Plantilla

```
#include <bits/stdc++.h>
using namespace std;

// ─── Fast IO
──────────────────────────────────────────────────────────────
──
#define ios ios::sync_with_stdio(false); cin.tie(nullptr);

// ─── Debug (solo activo en LOCAL, se elimina en juicio)
────────────────────
#ifdef LOCAL
  #define dbg(x)  cerr << "[" << #x << "] = " << (x) << "\n"
  #define dbgv(v) cerr << "[" << #v << "] = "; for(auto _x:
(v)) cerr<<_x<<" "; cerr<<"\n"
  #define dbgm(m) for(auto& _r:(m)){for(auto _x:_r)cerr<<_x<<"
";cerr<<"\n";}
#else
  #define dbg(x)
  #define dbgv(v)
  #define dbgm(m)
#endif

// ─── Macros de iteración
──────────────────────────────────────────────
#define rep(i,n)    for(int i=0;i<(int)(n);i++)
#define rep1(i,n)   for(int i=1;i<=(int)(n);i++)
#define repr(i,n)   for(int i=(int)(n)-1;i>=0;i--)
#define each(x,a)   for(auto& x:(a))

// ─── Macros de contenedor
─────────────────────────────────────────────
#define all(s)   (s).begin(),(s).end()
#define rall(s)  (s).rbegin(),(s).rend()
#define pb       push_back
#define eb       emplace_back
#define sz(v)    ((int)((v).size()))

// ─── Shortcuts
──────────────────────────────────────────────────────────────
#define nl       '\n'
#define ff       first
#define ss       second
#define yesi     cout<<"Yes\n"
#define nosi     cout<<"No\n"

// ─── Typedefs
──────────────────────────────────────────────────────────────
──
typedef long long      ll;
typedef long double    ld;
typedef pair<int,int>  ii;
typedef pair<ll,ll>    pll;
typedef vector<int>    vi;
typedef vector<ll>     vll;
typedef vector<ld>     vld;
typedef vector<ii>     vii;
typedef vector<pll>    vpll;
typedef vector<vi>     vvi;
typedef vector<vll>    vvll;
typedef vector<string> vs;
typedef vector<char>   vc;

// ─── Constantes
──────────────────────────────────────────────────────────
const ll LINF = (1LL << 62);

// ─── ckmin / ckmax
──────────────────────────────────────────────────────────
template<class T> bool ckmin(T& a, const T& b){ return b<a ?
a=b,1:0; }
template<class T> bool ckmax(T& a, const T& b){ return b>a ?
a=b,1:0; }

// ─── Bit helper
──────────────────────────────────────────────────────────
int bit(int mask, int b){ return (mask >> b) & 1; }

// ─── RNG
──────────────────────────────────────────────────────────────
──────
mt19937
rng(chrono::steady_clock::now().time_since_epoch().count());
#define randint(a,b) uniform_int_distribution<int>(a,b)(rng)

// ─── Lectura rápida genérica
──────────────────────────────────────────────
```

```
template<class T>            void read(T& x)            { cin
>> x; }
template<class F, class S>   void read(pair<F,S>& p)    { cin
>> p.first >> p.second; }
template<class T>            void read(vector<T>& v)    {
each(x,v) read(x); }

//
------------------------------------------------------------------------
--------------

void solve() {

}

int main(){
    ios
    int t = 1;
    // cin >> t;
    while(t--){
      solve();
    }
    return 0;
}
```